UNIT 3

Memory & I/O Interfacing: Types of memory – Memory mapping and addressing – Concept of I/O map – types – I/O decode logic – Interfacing key switches and LEDs – 8279 Keyboard/Display Interface - 8255 Programmable Peripheral Interface – Concept of Serial Communication – 8251 USART – RS232C Interface.

1. Difference between I/O mapped I/O and Memory Mapped I/O technique. ***

Characteristics	Memory Mapped I/O technique	I/O mapped I/O (Peripheral I/O)
1. Device address	16- bit	8-bit
2. Control signals for Input/ Output	$\overline{MEMR}$/$\overline{MEMW}$	$\overline{IOR/IOW}$
3. Instructions available	Memory-related instructions such as STA; LDA; LDAX; STAX; MOV M,R; ADD M;SUB M; ANA M; etc.	IN and OUT
4. Data transfer	Between any register and I/O	Only between I/O and the accumulator
5. Maximum number of I/Os possible	The memory map (64K) is shared between I/Os and system memory	The I/O map is independent of the memory map; 256 input devices and 256 output devices can be connected.
6. Execution Speed	13 T-states (STA,LDA) 7 T-states (MOV M,R)	10 T-states
7. Hardware requirements	More hardware is needed to decode 16- bit address	Less hardware is needed to decode 8-bit address
8. Other features	Arithmetic or logical operations can be directly performed with I/O data	Not available

2. What are the functions of BSR mode? ***

The Bit Set/Reset mode is for port C only. In this mode there is no effect on the I/O mode.

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0

D_7=0
BSR flag is active

Not used

Port c bit selected
000-Bit 0
001-Bit 1
10- Bit 2
11- Bit 3
100- Bit 4
101- Bit 5
110- Bit 6
111- Bit 7

1 = Bit SET
0 = Bit RESET

3. Discuss about RS232 line drivers ***

- This is a quad line driver that converts TTL input to maximum 15V output
- It is used with the power supply ±12V
- Logic 0 input (<0.8 V) the output is around -10V and logic 1 input (>2.4 V) the output is around +10V.

4. List the features of USART ***

- The **Universal Synchronous/Asynchronous Receiver/Transmitter** is a serial data communication
- The Intel 8251(USART) is the hardware device that converts parallel to serial data and vice-versa and overcomes such overburden of large program. Hence it is called Programmable Communication Interface.
- It is packaged in a 28-pin DIP
- Receives parallel data from the CPU & transmits serial data after conversion.
- It supports both synchronous and asynchronous modes of operation
- The synchronous baud rate — DC to 64 K baud and asynchronous baud rate — DC to 19.2 K baud

5. What are the types of memory mapping and list their advantages and disadvantage

- **Memory mapped I/O**
  - ✓ No separate IO/M
  - × Memory space is limited
- **I/O Mapped I/O**
  - ✓ More space for memory
  - × Only 2 instruction are available for I/O access.

6. What are the modes of operation supported by 8255?

- Simple I/O mode
- Strobed I/O mode
- Bidirectional mode.

7. List the functions performed by 8279?***

- Keyboard Scanning
- Key debouncing
- Keycode generation
- Informing the key entry to CPU.
- Storing display codes
- Output display codes to LED's
- Display refreshing.

8. What is stack and subroutine?***

***** The stack is a group of memory locations in the R/W memory that is used for temporary storage of binary information during the execution of the program.

* Subroutine is a group of instructions written separately from the main program to perform a function that occurs repeatedly in the main program.

Note: After the completion of the subroutine task, the microprocessor returns to the main program. The subroutine technique eliminates the need to write a subtask repeatedly; thus, it uses memory more efficiently. Before executing the subroutine technique, the stack must be defined; the stack is used to store the memory address of the instruction the main program that follows the subroutine call.

9. Differentiate absolute decoding and partial decoding concepts used in I/O decode logic of 8085 microprocessor.***

Absolute decoding	Partial decoding
All eight address lines are decoded to generate one unique output pulse; the device will be selected only with the address, $01_{H.}$ this is called absolute decoding and is a good design practice It is used in large systems.	To minimize the cost, the output port can be selected by decoding some of the address lines, this is called partial decoding. A_1 and A_0 are not connected, and they are replaced by IO/M and WR signals. It is used in small systems.

10. List the salient features of Mode0 of 8255?***

- Two 8 bit ports (Port A and Port B) and two 4 bit ports (Port C upper and lower) are available.
- The two 4 bit ports can be combinedly used as third 8 bit port.
- Any port can be used as an input or output port.
- Output ports are latched. Input ports are not latched.

- A maximum of 4 ports are available so that overall 16 I / O configurations are possible.

11. What is key bouncing?

Mechanical switches are used as keys in most of the keyboards. When a key ispressed the contact bounce back and forth and settle down only after a small timedelay (about 20ms). Even though a key is actuated once, it will appear to havebeen actuated several times. This problem is called Key Bouncing.

12. What is an USART?***

USART stands for universal synchronous/Asynchronous Receiver/Transmitter. It is a programmable communication interface that can communicate by using either synchronous or asynchronous serial data.

13. What is the output modes used in 8279?

8279 provides two output modes for selecting the display options.

- **Display Scan:**In this mode, 8279 provides 8 or 16 character-multiplexed displays those can beorganized as dual 4-bit or single 8-bit display units.
- **Display Entry:**8279 allows options for data entry on the displays. The display data is entered fordisplay from the right side or from the left side.

14. What are the modes used in keyboard modes?**

- Scanned Keyboard mode with 2 Key Lockout.
- Scanned Keyboard with N-key Rollover.
- Scanned Keyboard special Error Mode.
- Sensor Matrix Mode.

15. What is TXD?*

TXD- Transmitter Data Output
This output pin carries serial stream of the transmitted data bits along with other information likestart bit, stop bits and priority bit.

16. What is RXD?*

RXD- Receive Data Input
This input pin of 8251A receives a composite stream of the data to be received by 8251A.

17. What is the use of modem control unit in 8251?*

The modem control unit handles the modem handshake signals to coordinate thecommunication between the modem and the USART.

18. What is the purpose of 8255 PPI?

The 8255A is widely used, programmable, parallel I/O device .It canbe programmed to transfer data under various conditions, from simple I/O tointerrupt I/O.

19. Define Baud?

The rate at which the bits are transmitted is called Baud.

20. Write an instruction for serial output data?

MVI A, 80H : Set D7 in the accumulator=1
RAR ; Set D6 =1
SIM

21. Define serial to parallel conversion?**

In serial reception, the MPU receives a stream of eight bits andit is converted in to 8-bit parallel word. This is known as serial to parallelconversion.

22. Define parallel to serial conversion?

In serial transmission an 8-bit parallel word should be converted into a stream of eight serial bits. This is known as parallel to serial conversion.

UNIT III

Memory & I/O Interfacing: Types of memory – Memory mapping and addressing – Concept of I/O map – types – I/O decode logic – Interfacing key switches and LEDs – 8279 Keyboard/Display Interface - 8255 Programmable Peripheral Interface – Concept of Serial Communication – 8251USART – RS232C Interface.

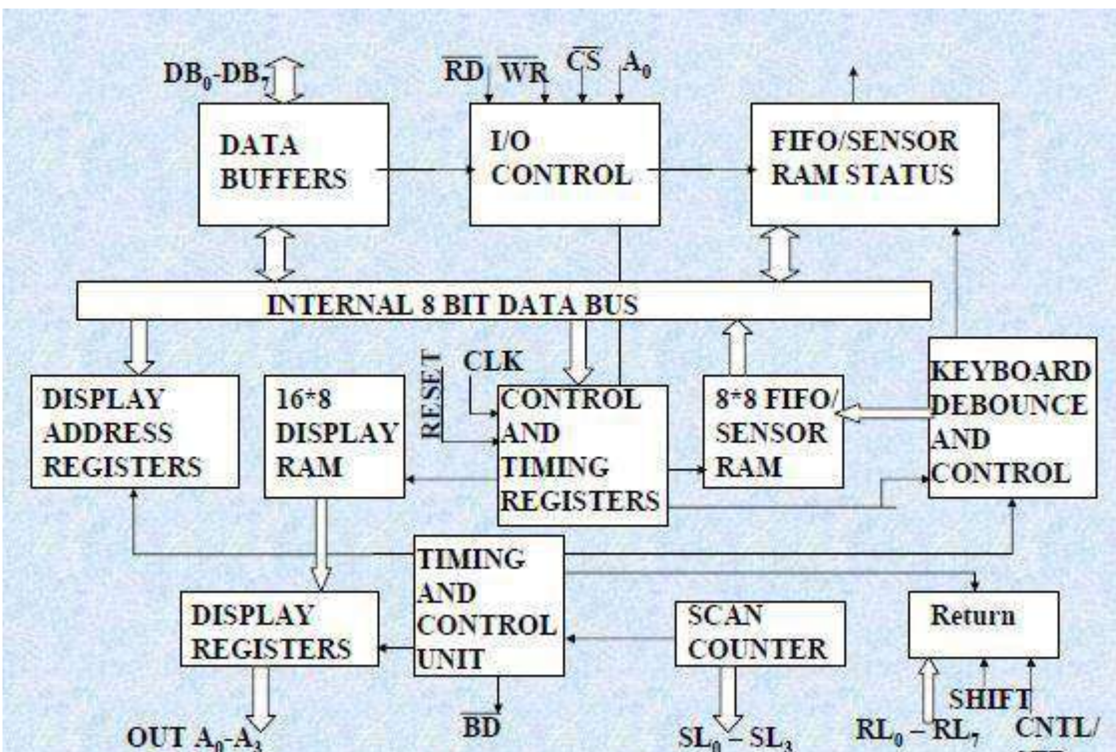
8279 Keyboard/Display Interface

PROGRAMMABLE KEYBOARD/DISPLAY INTERFACE - 8279

- The 8279 is a hardware approach to interfacing a matrix keyboard and multiplexed display.
- It is a 40 - pin device with 2 major segments: (i) keyboard and (ii) Display

BLOCK DIAGRAM

- Four major sections are
    1. Keyboard section
    2. Scan section
    3. Display section
    4. MPU interface section.



8279 Internal Architecture

KEYBOARD SECTION

- Eight lines (RL0-RL7) connected to 8 columns of keyboard & two additional lines **shift &control/strobe** keys are automatically debounced and the keyboard can operate in two modes.

(i) Two key lockout mode

If two keys are pressed almost simultaneously, only the 1st key is recognized.

(ii) N-key rollover mode

If one or more keys are pressed simultaneously, each keys are recognized and their codes are stored in the internal buffer; it can also setup so that no key is recognized until only one key remain pressed.

- Keyboard section also includes 8×8 FIFO RAM.
- FIFO RAM consists of 8 register that can store 8 keyboard entries, each is then read in the order of entries.
- FIFO RAM STATUS used to check the status of FIFO RAM, that is it contain any character to be displayed; if supposed means it provide interrupt request through IRQ.

SCAN SECTION

- Four scan lines (SL0-SL3)
- Four scan lines can be decoded using a 4 to 16 decoder to generate 16 lines for scanning.
- Four lines can be connected to the rows of a matrix keyboard & the digit diverts of a multiplexed display.

DISPLAY SECTION

- It has 8 output lines divided into 2 groups A_0-A_3 and B_0-B_3.
- It can be used either 2 group of 8 lines or group of 8 lines.
- The display can be blanked by using the BD line.
- This section includes 16 × 8 display RAM.

MPU INTERFACE SECTION

The signal description of each of the pins of 8279 as follows:

DB_0-DB_7: These are bidirectional data bus lines. The data and command words to and from the CPU are transferred on these lines.

CLK: This is a clock input used to generate internal timing required by 8279.

RESET: This pin is used to reset 8279. A high on this line reset 8279. After resetting 8279, it's in sixteen 8-bit display, left entry encoded scan, 2-key lock out mode. The clock prescaler is set to 31.

CS: Chip Select – A low on this line enables 8279 for normal read or write operations.

Otherwise, this pin should remain high.

A0: A high on this line indicates the transfer of a command or status information. A low on this linen indicates the transfer of data. This is used to select one of the internal registers of 8279.

RD, WR (Input/output) READ/WRITE – These input pins enable the data buffers to receive or send data over the data bus.

IRQ: This interrupt output lines goes high when there is a data in the FIFO sensor RAM. The interrupt lines goes low with each FIFO RAM read operation but if the FIFO RAM further contains any key-code entry to be read by the CPU, this pin again goes high to generate an interrupt to the CPU.

Vss, Vcc: These are the ground and power supply lines for the circuit.

8279 INTERFACE WITH 8085

- Interface with 8279 with 8085 using I/O mapped I/O technique.
- Let us take address D0 to select Data Register & D1 to select Command Register.
- An 8-input NAND gate is used to select the chip 8279.
- During IN and OUT instruction, the port address is available on A0 to A7 as well as on A8 – A15.
- The 8279 interface shown in figures the address line A0-A7.
- The address line A0 of 8085 is directly connected to A0 of 8279.
- The remaining address A1 to A7 are used with the 8-input NAND gate to get a ‘0’ at the output of the NAND gate which is used to select 8279.

A7	A6	A5	A4	A3	A2	A1	A0	
1	1	0	1	0	0	0	0	$D0_H$
1	1	0	1	0	0	0	1	$D1_H$

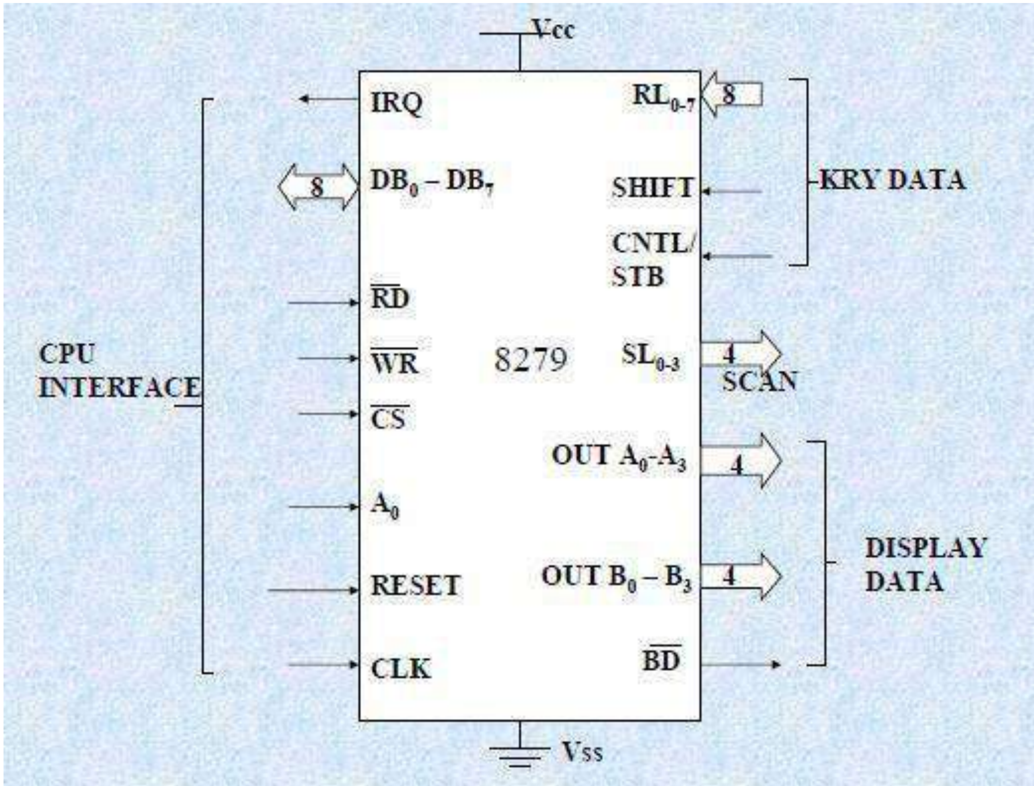
The address bits A7, A6 & A4 are connected to the NAND gate.

8279 can also be interfaced to 8085 using memory mapped I/O technique.

SL0-SL3-Scan Lines: These lines are used to scan the key board matrix and display digits. These lines can be programmed as encoded or decoded, using the mode control register.

RL0 - RL7 - Return Lines: These are the input lines which are connected to one terminal of keys, while the other terminal of the keys are connected to the decoded scan lines. These are normally high, but pulled low when a key is pressed.

SHIFT: The status of the shift input lines is stored along with each key code in FIFO, in scanned keyboard mode. It is pulled up internally to keep it high, till it is pulled low with a key closure.



BD – Blank Display: This output pin is used to blank the display during digit switching or by a blanking closure.

OUT A0 – OUT A3 and OUT B0 – OUT B3 – These are the output ports for two 16*4 or 16*8 internal display refresh registers. The data from these lines is synchronized with the scan lines to scan the display and keyboard. The two 4-bit ports may also as one 8-bit port.

CNTL/STB- CONTROL/STROBED I/P Mode: In keyboard mode, this line is used as a control input and stored in FIFO on a key closure. The line is a strobed line that enters the data into FIFO RAM, in strobed input mode. It has an interrupt pull up. The lines are pulled down with a key closer.

CONTROL WORD TO SET KEYBOARD-DISPLAY MODE

D D

0 0→ Eight 8- bit character display- left entry

0 1→Sixteen 8-bit character display- left entry

1 0→ Eight 8- bit character display- Right entry

1 1→ Sixteen 8- bit character display- Right entry

K K K

0 0 0 →encoded scan keyboard- 2key lockout

0 0 1→decoded scan keyboard – 2 key lockout

0 1 0→encoded scan keyboard- N key rollover

0 1 1→ decoded scan keyboard- N key rollover

1 0 0→encoded scan sensor matrix

1 0 1→ decoded scan sensor matrix

1 1 0→strobed input, encoded display scan

1 1 1→ strobed input, decoded display scan

Keyboard mode

Encoded: 4-bit output from 16-bit input 0000 to 1111

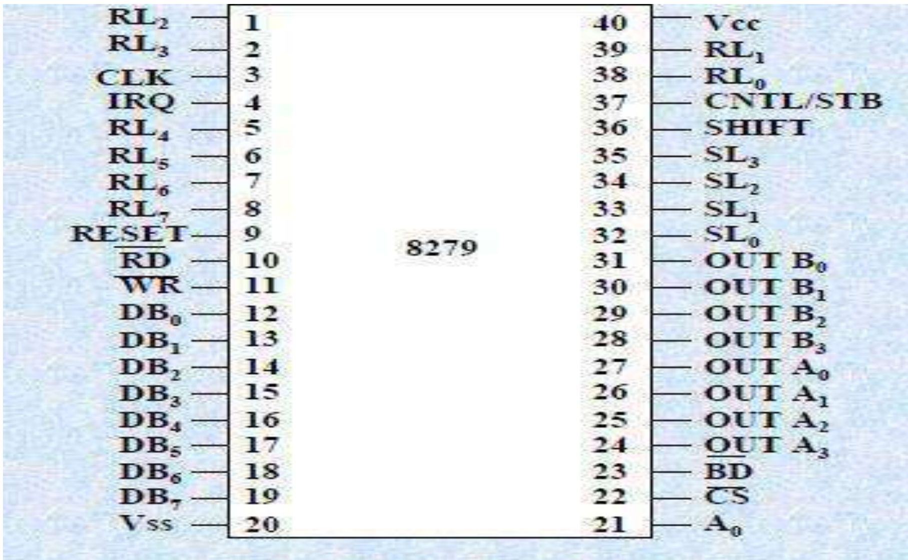
Decoded: input 4 to 16-bit decoder

Control word to write into display RAM

1	**0**	**0**	**A**	**A**	**A**	**A**	**A**

Control word to clear display

1	**1**	**0**	**CD**	**CD**	**CD**	**CF**	**CA**

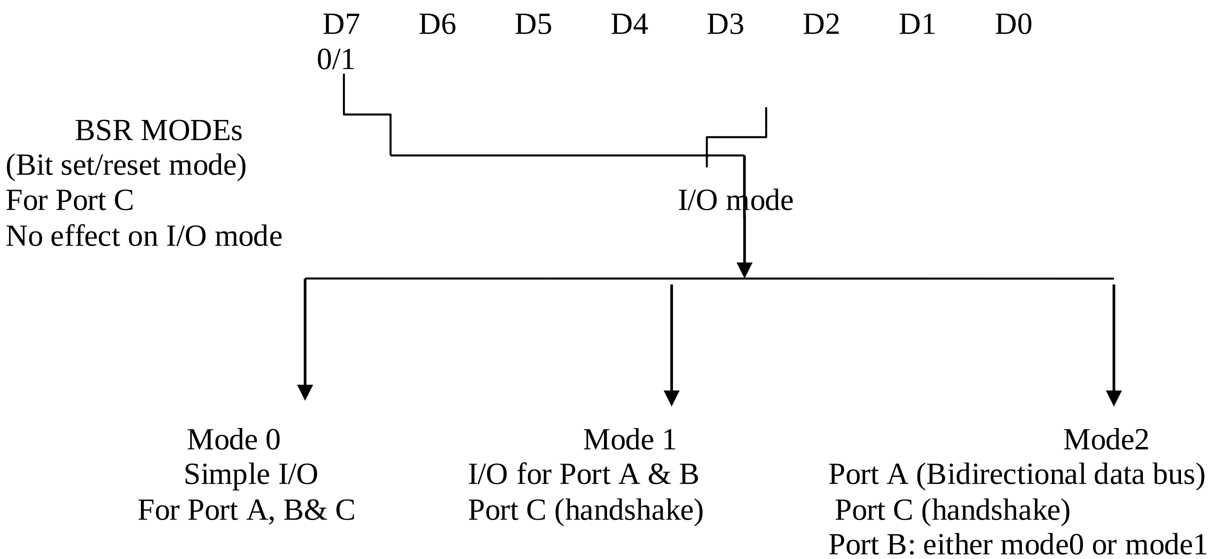


8279 Pin Configuration

8255 Programmable Peripheral Interface

- 8255 device consists of 4 ports, for parallel I/O.
- It is a 40-pin IC.
- 28-pin is used for I/O Ports
- Port A→ 8-bis
- Port B→8-bits
- Port C(upper)→4-bits
- Potr C(lower)→4-bits

Control word



BLOCK DIAGRAM 0F 8255

1. **Control logic**

The control section consists of six lines.

Read ($\overline{RD}$): this control signal enables the read operation. When the signal low the MPU reads data from a selected I/O port of 8255.

Write ($\overline{WR}$): this control signal enables the write operation. When the signal low the MPU writes data into a selected I/O port of 8255.

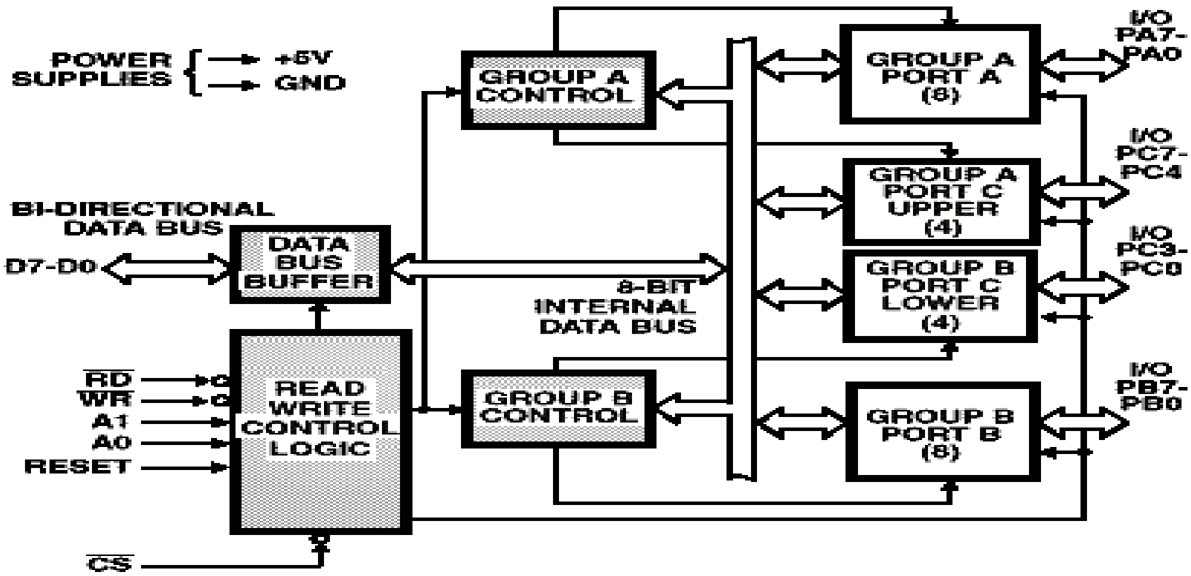
RESET (Reset): clears the control register & sets all port in I/O mode.

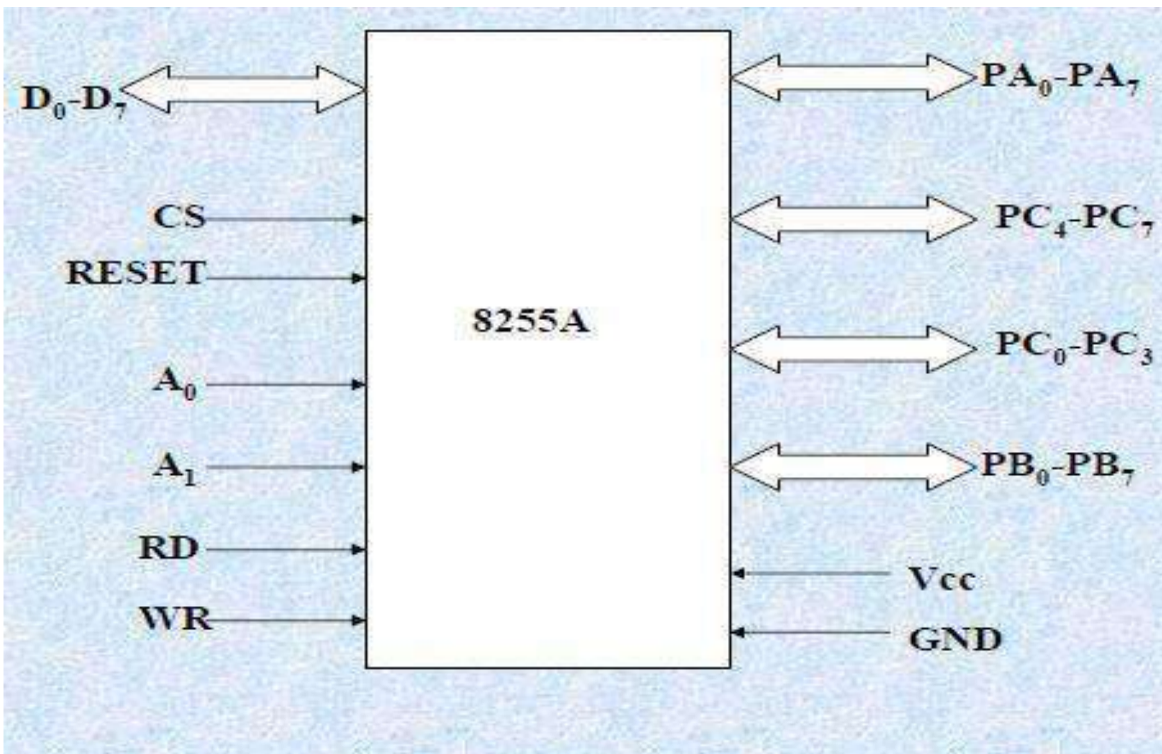
CS, A0 & A1: These are device select signals. CS is connected to a decoded address and A0 & A1 are generally connected to MPu address line A0 & A1.

CS-> master chip select

A0 & A1-> specify one of the ports

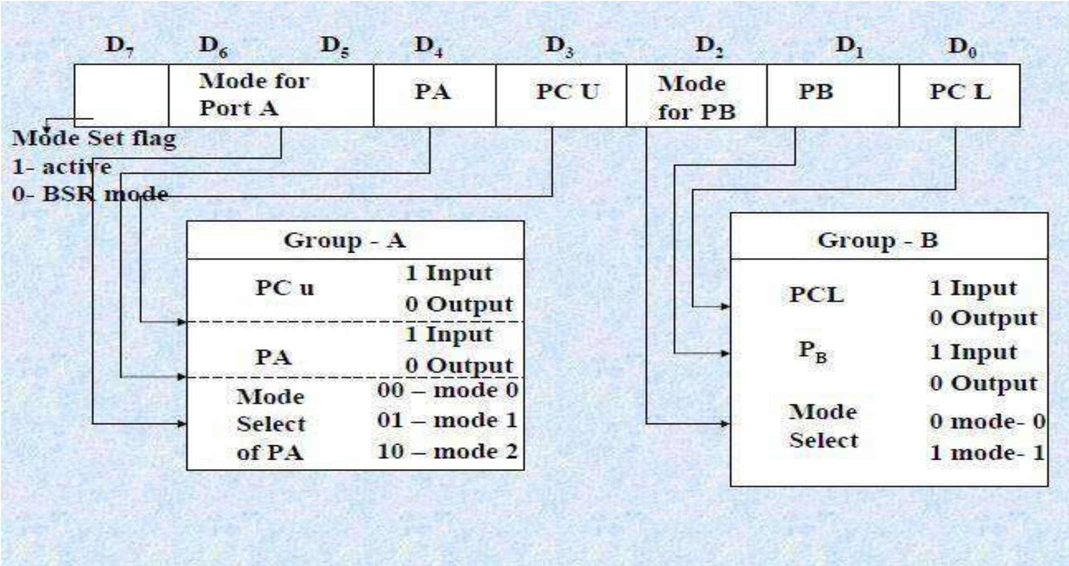
CS	**A0**	**A1**	**Hex. Address**	**selected**
0	0	0	80H	Port A
0	0	1	81H	Port B
0	1	0	82H	Port C
0	1	1	83H	control register
1	×	×	-	8255 is not selected





Signals of 8255

Control word format



Control Word Format of 8255

MODES OF 8255

Mode 0: Simple input or output

In this mode, port A and B are used as two simple 8-bit I/O ports & port C as two 4 bits. The input/output features of mode0 are as follows:

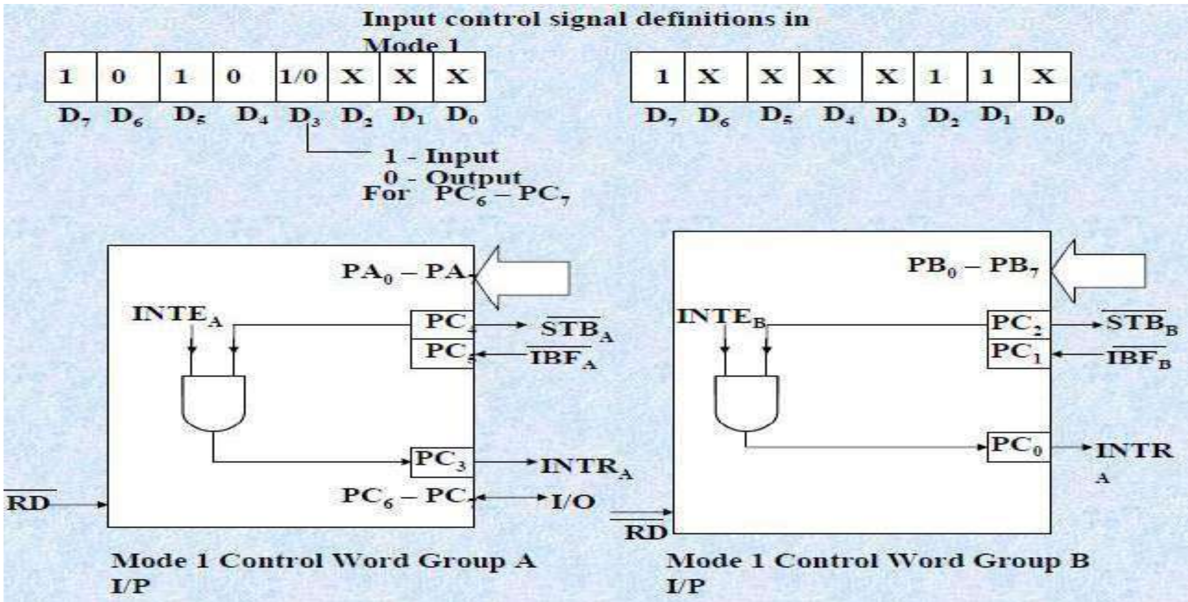
1. Outputs are latched
2. Inputs are no latched
3. Ports do not have handshake or interrupt capability.

Mode 1: input or output with handshake

In this mode, data is transferred between MPU & peripheral using handshake signal.

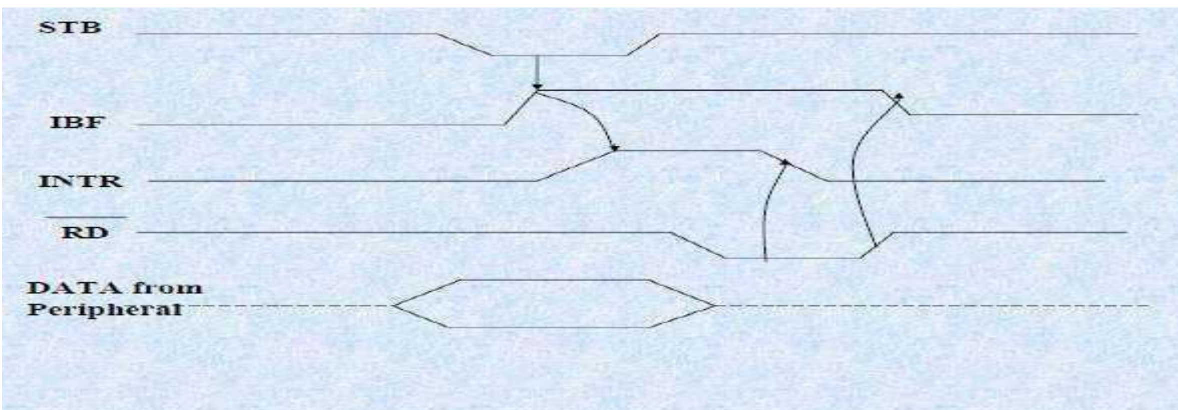
The features of this mode include

- Two ports (A&B) function as 8-bit I/O ports.
- Each port uses three lines from port C as handshake signals.
- The remaining two signals of port C can be used for simple I/O functions.
- Input or output data are latched.
- Interrupted logic is supported.

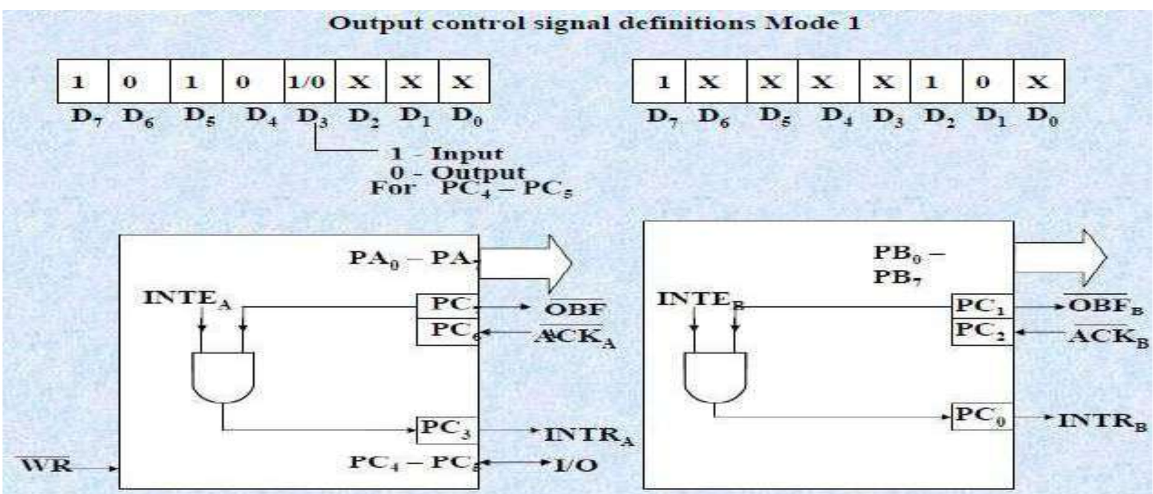


Mode 1 Control Word Group A I/P

Mode 1 Control Word Group B I/P

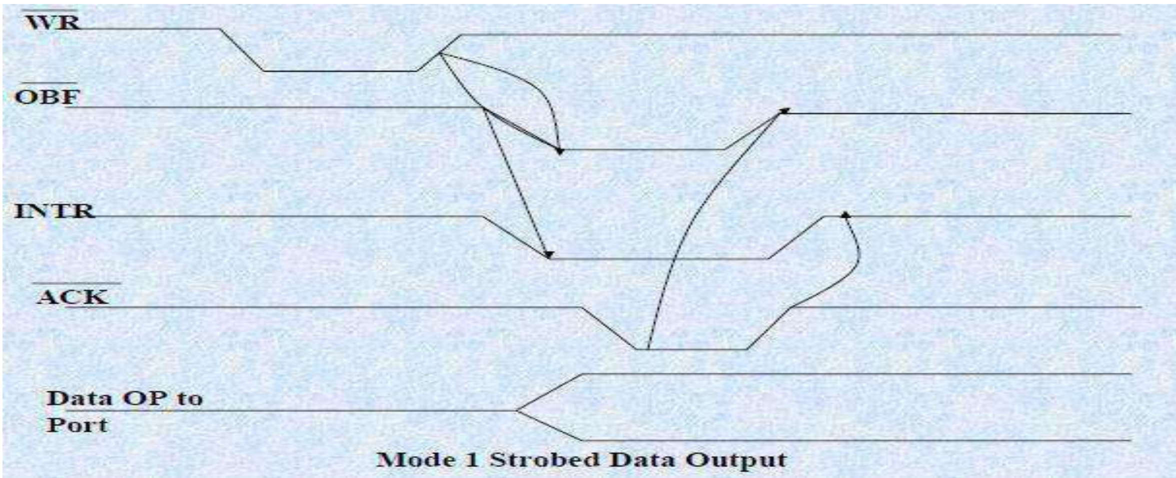


Mode 1 Strobed Input Data Transfer



Mode 1 Control Word Group A

Mode 1 Control Word Group B

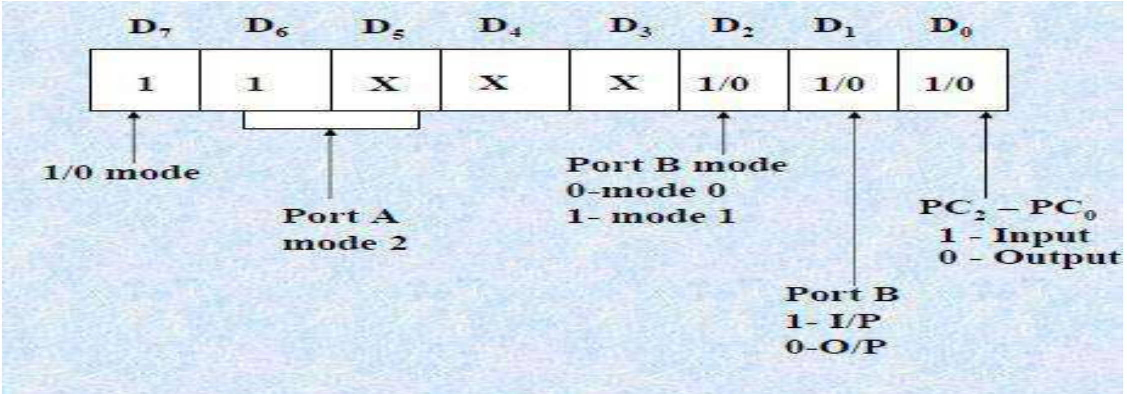


MODE 2 *(STROBED BIDIRECTIONAL I/O):*

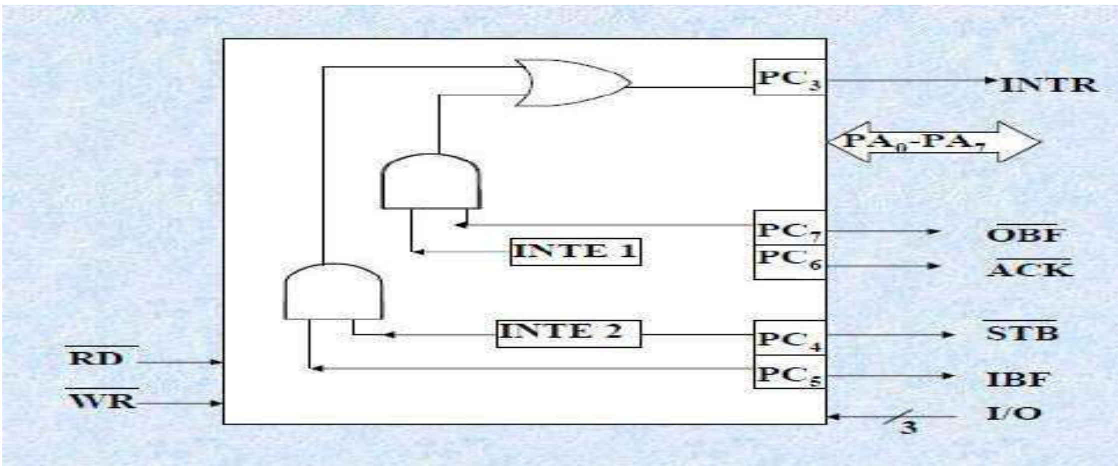
- This mode of operation of 8255 is also called as strobed bidirectional I/O.
- This mode of operation provides 8255 with additional features for communicating with a peripheral device on an 8-bit data bus.
- Handshaking signals are provided to maintain proper data flow and synchronization between the data transmitter and receiver.
- The interrupt generation and other functions are similar to mode 1.
- In this mode, 8255 is a bidirectional 8-bit port with handshake signals.
- The RD and WR signals decide whether the 8255 is going to operate as an input port or output port.

The Salient features of Mode 2 of 8255 are listed as follows:

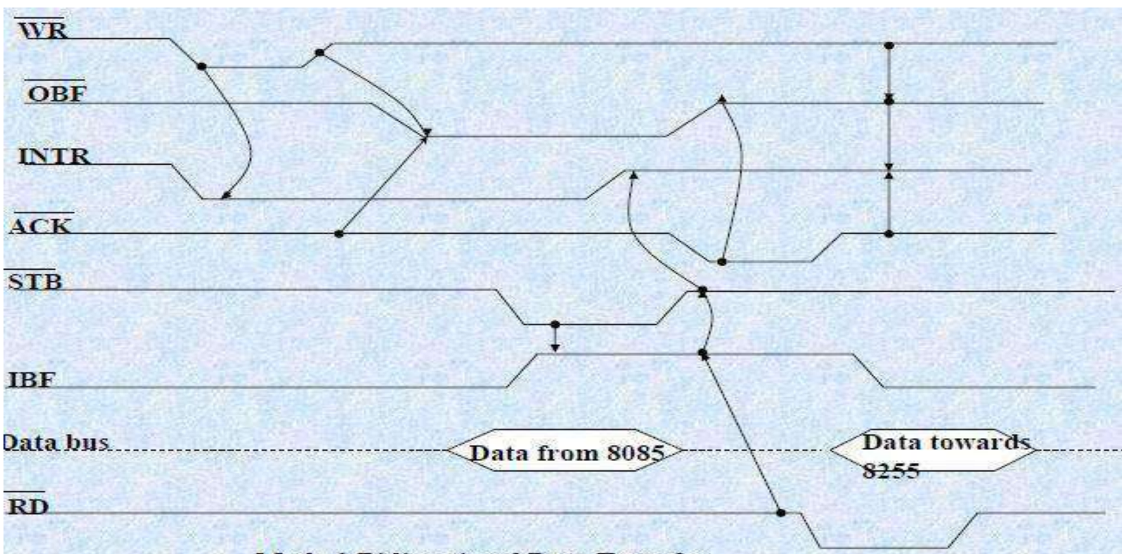
1. The single 8-bit port in group A is available.
2. The 8-bit port is bidirectional and additionally a 5-bit Control port is available.
3. Three I/O lines are available at port C. (PC2 – PC0)
4. Inputs and outputs are both latched.
5. The 5-bit control port C (PC3-PC7) is used for generating / accepting handshake signals for the 8-bit data transfer on port A.



Mode 2 control word



Mode 2 pins



Mode 2 Bidirectional Data Transfer

Control signal definitions in mode 2:

- **INTR** – (Interrupt request) As in mode 1, this control signal is active high and is used to interrupt the microprocessor to ask for transfer of the next data byte to/from it. This signal is used for input (read) as well as output (write) operations.

Control Signals for Output operations:

- **OBF** (Output buffer full) – This signal, when falls to low level, indicates that the CPU has written data to port A.
- **ACK** (Acknowledge) : This control input, when falls to logic low level, acknowledges that the previous data byte is received by the destination and next byte may be sent bythe processor. This signal enables the internal tristate buffers to send the next data byte on port A.
- **INTE1** (A flag associated with OBF) This can be controlled by bit set/reset mode with PC6.

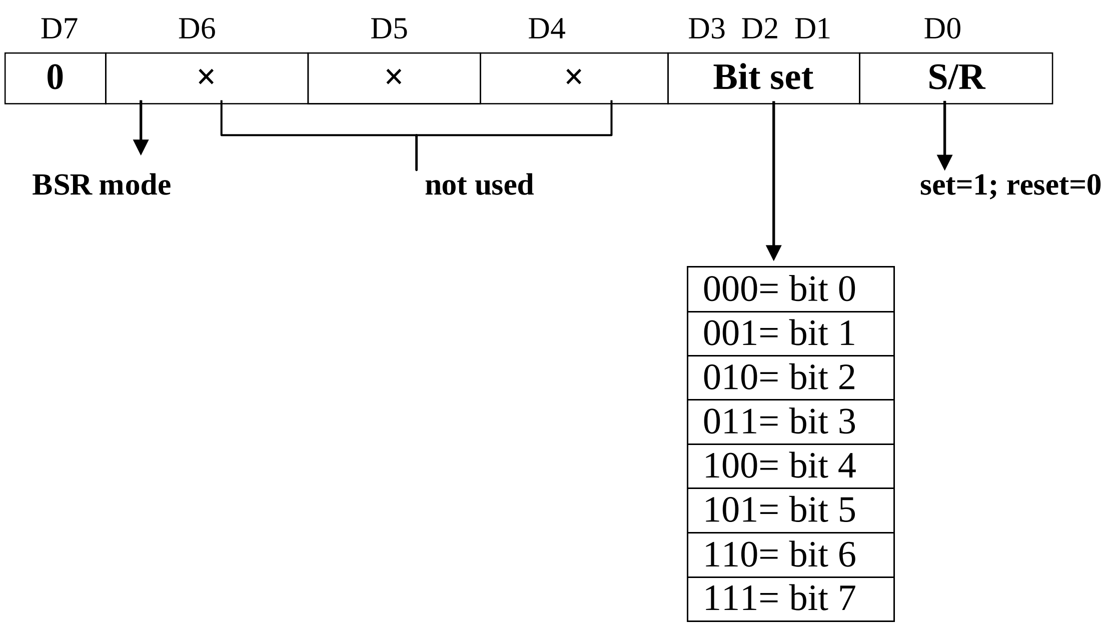
Control signals for input operations:

- **STB** (Strobe input): A low on this line is used to strobe in the data into the input latches of 8255.
- **IBF** (Input buffer full): When the data is loaded into input buffer, this signal rises to logic ‘1’. This can be used as an acknowledge that the data has been received by the receiver.
- The waveforms in fig show the operation in Mode 2 for output as well as input port.

BSR (BIT SE RESET MODE)

The BSR mode is concerned only with bits of port C, which is either se or reset by writing appropriate control word in the control register.

BSR control word format



Explain in detail about 8251A USART

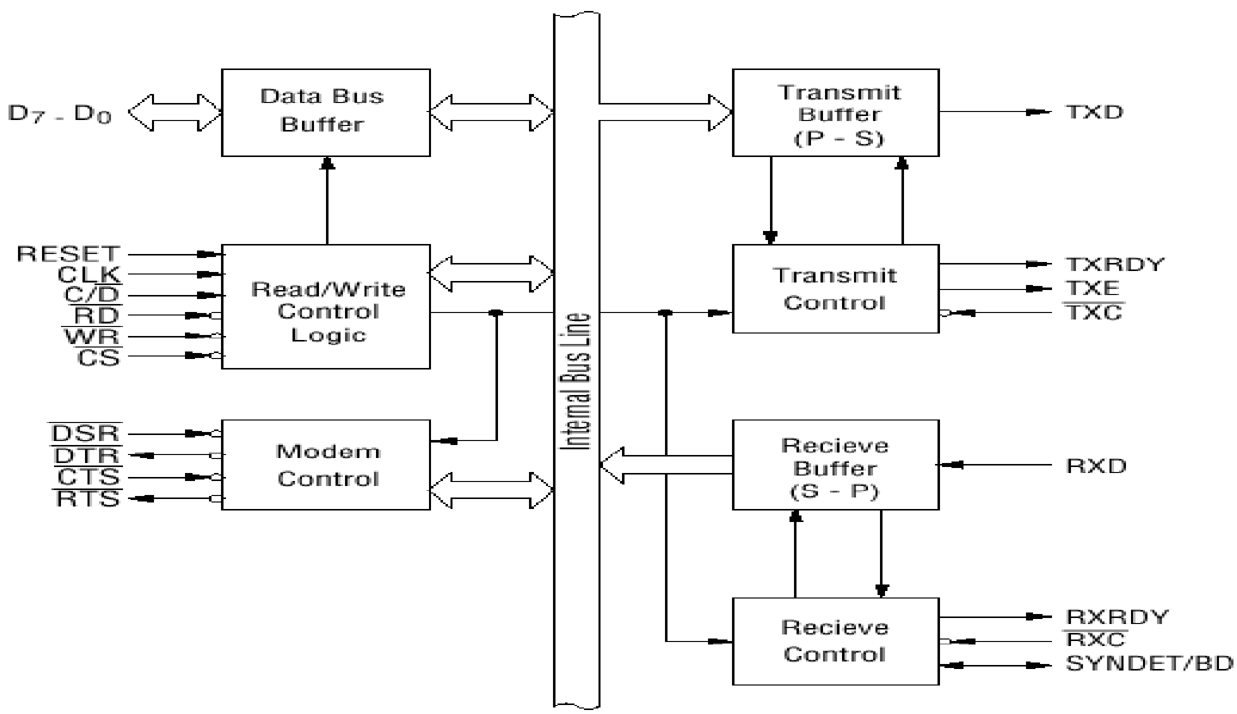
Featuresof8251

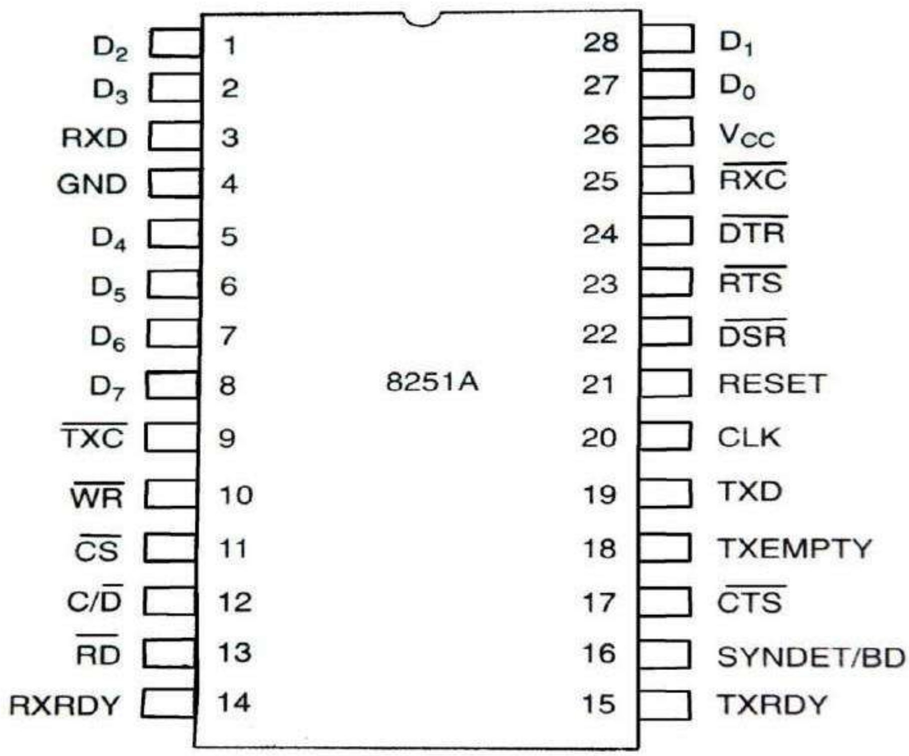
8251A is a USART - serial data communication.

USART-UNIVERSAL SYNCHRONOUS/ASYNCHRONOUS RECEIVER/TRANSMITTR

- packaged in a 28-pin DIP.
- Receives parallel data from the CPU & transmits serial data after conversion.
- Also receives serial data from the outside & transmits parallel data to the CPU after conversion.
- It supports both synchronous and asynchronous modes of operation.
- The synchronous baud rate — DC to 64 K baud.
- The asynchronous baud rate — DC to 19.2 K baud.
- In synchronous mode & asynchronous mode it supports 5-8 bits characters.
- supports both synchronous as well as asynchronous data transfer
- It contains full duplex
- double buffered system
- It provides error detection logic, detects parity over run, framing errors
- It uses separate TxC and RxC clock inputs for transmitter and receiver
- So transmitter and receiver can be operated in different baud rates

Pin diagram & block diagram of 8251A





Block diagram of 8251A consists of 5 sections

- Data Bus buffer
- Read/Write Control Logic
- Modem Control
- Transmitter
- Receiver

Data Bus buffer

- D0-D7 : It is a 3 state bi-directional transfers control word, status word (transmitter & receiver) from R/W control logic

Read/Write Control Logic

- Includes control logic, six input signals & three buffer registers: Data register, control register & status register.
- Control logic: Interfaces the chip with MPU, determines the functions of the chip according to the control word in the control register & monitors the data flow.
- Control signals such as RD, WR C/D CS, CLK and Reset (control bus and control signals)

INPUT SIGNALS

- CS – Chip Select : When signal goes low, the 8251A is selected by the MPU for communication.
- C/D – Control/Data : When signal is high, the control or status register is addressed; when it is low, data buffer is addressed. (Control register & status register are differentiated by WR and RD signals)
- WR : When signal is low, the MPU either writes in the control register or sends output to the data buffer.
- RD : When signal goes low, the MPU either reads a status from the status register or accepts data from data buffer.
- RESET : A high on this signal reset 8252A & forces it into the idle mode.
  CLK : Clock input, usually connected to the system clock for communication with the microprocessor.

CONTROL REGISTER- Maintains three register(data register, status register & data buffer)

- 16-bit register for a control word
- consist of two independent bytes

Mode word:

- Specifies the general characteristics of operation such as baud, parity, number of bits etc.

Command word:

Enables the data transmission and reception.
Register can be accessed as an output port when the Control/Data pin is high.

DATA REGISTER

Used as an input and output port when the C/D is low

CS	C/D	WR	RD	Operation
0	0	1	0	MPU reads data from data buffer
0	0	0	1	MPU writes data from data buffer
0	1	0	1	MPU writes a word to control register
0	1	1	0	MPU reads a word from status register
1	×	×	×	Chip is not selected for any operation

STAUS REGISTER

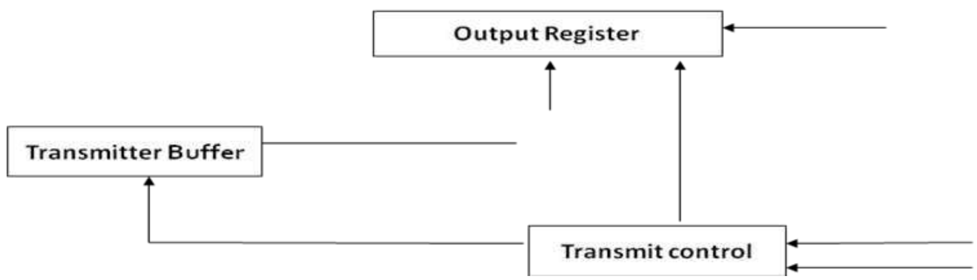
- Checks the ready status of the peripheral.
- Status word in the status register provides the information concerning register status and transmission errors.

Modem Control

- DSR - Data Set Ready : Checks if the Data Set is ready when communicating with a modem.
- DTR - Data Terminal Ready : Indicates that the device is ready to accept data when the 8251 is communicating with a modem.
- CTS - Clear to Send : If its low, the 8251A is enabled to transmit the serial data provided the enable bit in the command byte is set to '1'.
- RTS - Request to Send Data : Low signal indicates the modem that the receiver is ready to receive a data byte from the modem.

Transmitter

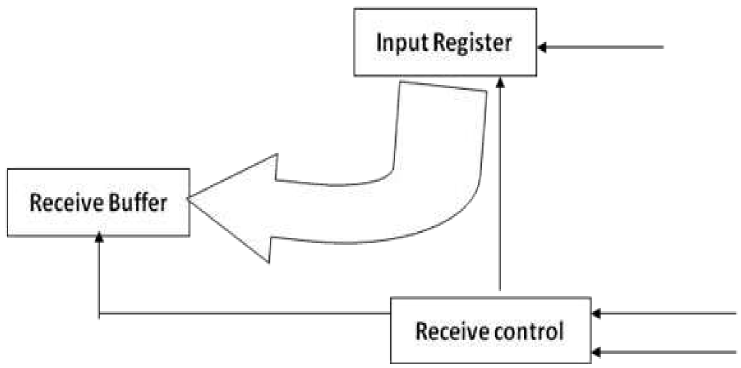
- Accepts parallel data from MPU & converts them into serial data.
- Has two registers:
  - Buffer register : To hold eight bits
  - Output register : To convert eight bits into a stream of serial bits



- The MPU writes a byte in the buffer register.
- Whenever the output register is empty; the contents of buffer register are transferred to output register.
- Transmitter section consists of three output & one input signals
  - TxD - Transmitted Data Output : Output signal to transmit the data to peripherals
  - TxC - Transmitter Clock Input : Input signal, controls the rate of transmission.
  - TxRDY - Transmitter Ready : Output signal, indicates the buffer register is empty and the USART is ready to accept the next data byte.
  - TxE - Transmitter Empty : Output signal to indicate the output register is empty and the USART is ready to accept the next data byte.

RECEIVER

- Accepts serial data on the RxD pin and converts them to parallel data.
- Has two registers :
  - Receiver input register
  - Buffer register



- When RxD goes low, the control logic assumes it is a start bit, waits for half bit time, and samples the line again. If the line is still low, the input register accepts the following data, and loads it into buffer register at the rate determined by the receiver clock.
- RxRDY - Receiver Ready Output: Output signal, goes high when the USART has a character in the buffer register & is ready to transfer it to the MPU.
- RxD - Receive Data Input : Bits are received serially on this line & converted into a parallel byte in the receiver input register.
- RxC - Receiver Clock Input : Clock signal that controls the rate at which bits are received by the USART.

RS-232C

Features:

- Data communication standard that defines the voltage in reference to 2 equipments, Data communication equipment (DCE) & Data terminal equipment (DTE)
- 3 signals => T_XD, R_XD & Ground.
- Terminals connected as DTE & DCE.
- Data transmitted over T_XD(8251) line are TTL logic level

- Transmitted bits are converted to RS-232 voltage levels and negative logic by time drivers (MC 1488).
- Data received by 8251 over R_XD line should be at TTL logic
- RS-232 signal in converted to positive TTL logic level by MC 1489.

Line Drivers (MC 1488):

- This is a quad line driver that converts TTL input to maximum 15V output.
- It is used with the power supply ±12V
- Logic 0 input (<0.8 V) the output is around -10V.
- Logic 1 input (>2.4 V) the output is around +10V.
- Positive true logic is converted into true logic for Rs 232 signals.
- It functions like a comparator.
    - Input lower than the threshold voltage and for the output is positive power supply voltage and for the input higher the threshold voltage, the output approximate negative power supply.

Line Receiver (MC 14889):

- This is quad line receiver it converts high voltage signal (+15V) into TTL logic level.
- The output voltage usually ranges from 0.2V (low) to 4.0V (high).
- When the transistor base has the negative input voltage, the transistor is turned off and the collector voltage (output of MC 1489) is high.
- When the transistor base has a positive input voltage, the transistor is driven into the saturation to 0.2V.

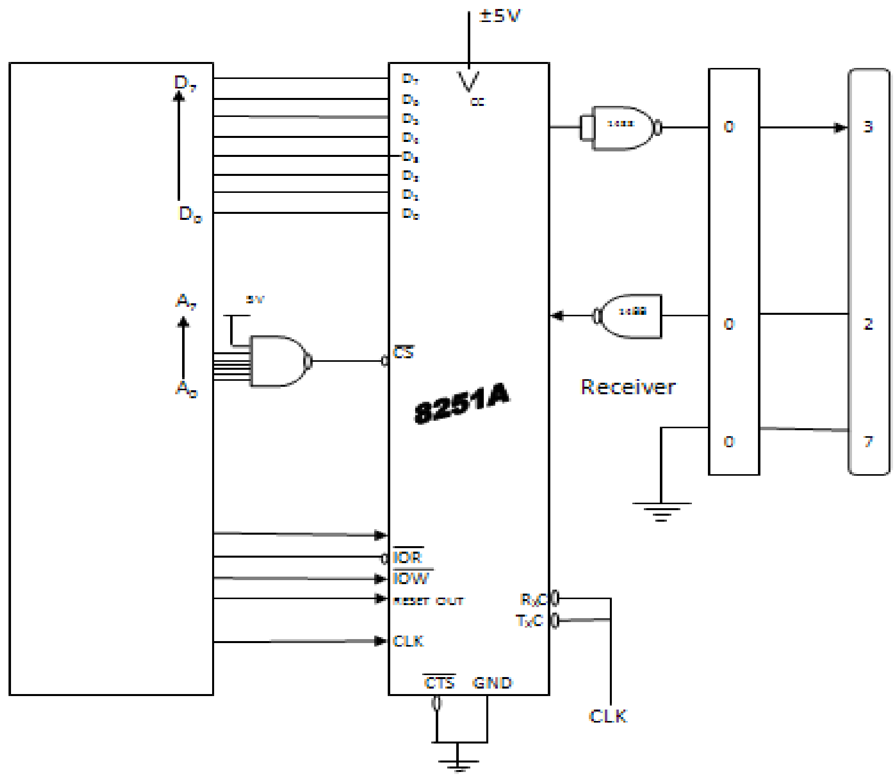
RS 232 Drivers/Receivers:

Max 232

- Drawback on the line driver Mc 1488 is that it requires additional ±12V from the power supply.
- This voltage is unavailable in system compactable for TTL.
- This is resolved by MC 232.
- It includes receiver and drivers, generate ±10V internally by using voltage doubler/inverter circuits.
- Max 232 can replace the 1488 and 1489 and eliminate the need if ±12V power supply.

RS-232

Schematic of Interfacing an RS-232 Terminal with am 8085 system using the 8251 A



Unit III

Memory & I/O Interfacing: Types of memory – Memory mapping and addressing – Concept of I/O map – types – I/O decode logic – Interfacing key switches and LEDs – 8279 Keyboard/Display Interface - 8255 Programmable Peripheral Interface – Concept of Serial Communication – 8251 USART – RS232C Interface.

MEMORY & I/O INTERFACING

Definition of Memory

Memory in a microprocessor system is where information (data and instructions) is kept.

1. TYPES OF MEMORY:

It can be classified into two main types:

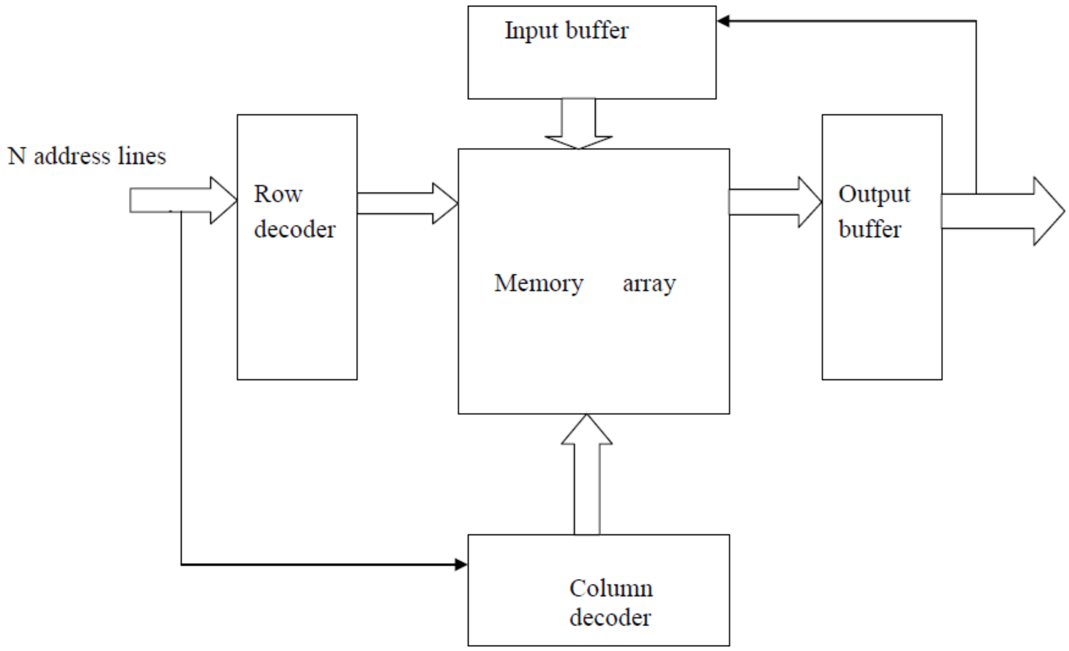
- Main memory (RAM and ROM)
- Storage memory (Disks, CD ROMs, etc.)

The simple view of RAM is that it is made up of registers that are made up of flip-flops (or memory elements). The number of flip-flops in a "memory register" determines the size of the memory word.

ROM on the other hand uses diodes instead of the flip-flops to permanently hold the information.

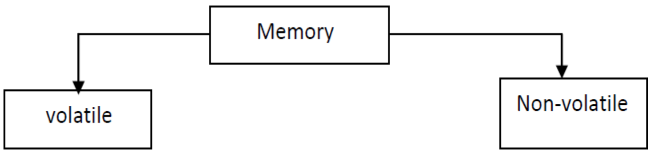
2. MEMORY MAP AND ADDRESSES

The memory map is a picture representation of the address range and shows where the different memory chips are located within the address range.



- The semiconductor memories alone have processor compatible access time for read and write operations. Therefore semiconductor memories are used as main memories

The different types of semiconductor memories are:



If the information stored in memory is lost when the power supply is switched OFF , then the memory is called volatile.

If the content of memory is preserved even after switching OFF the power supply then the memory is called non-volatile.

The volatile semi-conductor memories are static

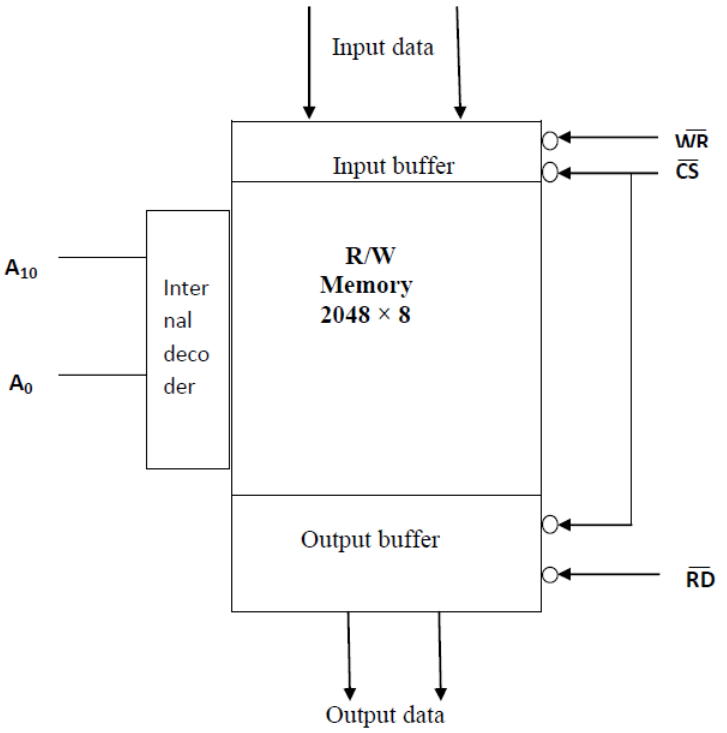
- Static RAM
- DRAM

The non-volatile semiconductor memories are

- ROM
- PROM
- EPROM
- NVROM

DRAM:

The DRAM's are read /write semiconductor memories in which the information is stored in the form of electric charge on the gate to substrate capacitance of a MOS –transistor.



Logic diagram for RAM

List the characteristics of DRAM

- The DRAMs are volatile and has random access feature.
- They are read/ write memories.
- The contents of DRAM have to be refreshed periodically using refreshing circuits
- The memory cell of DRAM will have 3-to-4 MOS-transistor.

Compare the static RAM and DRAM

Static RAM	DRAM
• Information is stored as voltage level in a flip-flop. • 6-to-8 transistors are required to form one memory cell. • Packing density is low. • The contents of memory need not be refreshed.	• Information is stored as a charge in the gate to substrate capacitance. • 3-to-4 transistors are required to form one memory cell. • Packing density is high. • The contents of memory have to be refreshed periodically.

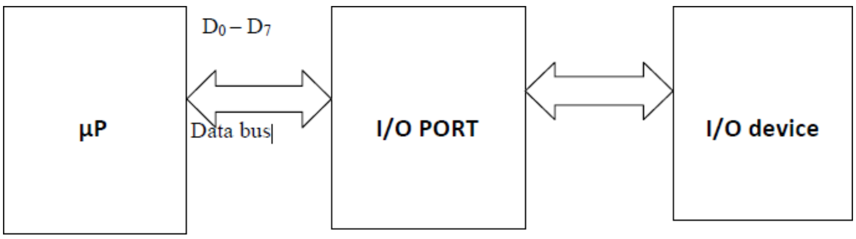
The chip select is the control signal that has to be asserted TRUE to bring an IC from high impedance state to normal state .Generally the chip select signals are generated in a system by decoding the unused address lines with the help of decoders .

Memory interfacing is done to access the memory quite frequently inorder to read instruction codes and data stored in memory. It deals with the following functions;

- choosing memories with suitable access time ,
- designing address decoding circuit to generate chip select signals,
- generating control signals for read/write operations and allocation of addresses to various memory IC's .

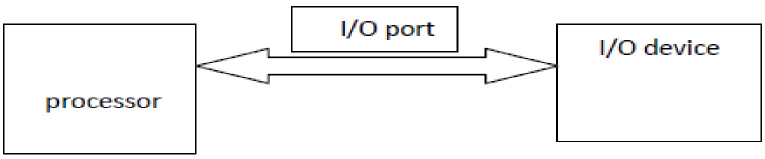
I/O interfacing

I/O interfacing is done using I/O ports. It is used to read/write data to /from the microprocessor and an I/O device. To read data from an input device an Input Port is used. To write data into an output device an Output Port is used.



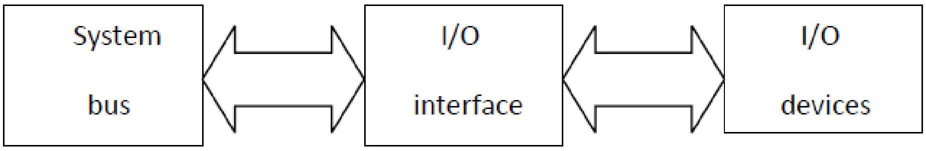
Programmed I/O

If the data transfer between an I/O device and the processor is accomplished through an I/O port and controlled by a program then the I/O device is called programmed I/o .



The need of interfacing in I/0 devices:

Generally I/O devices are slow devices. Therefore the speed of I/O devices does not match with the speed of microprocessor. And so an interface is provided between system bus and I/O devices.

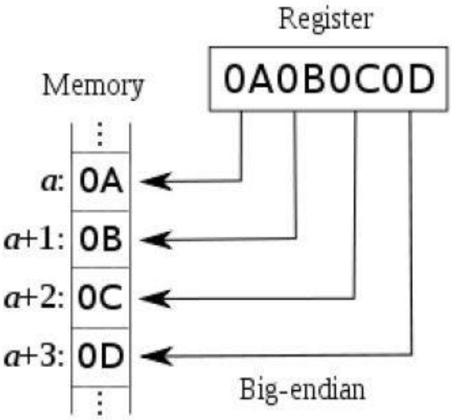


The difference between CPU bus and system bus:

The CPU bus has multiplexed lines but the system bus has separate lines for each signal. (The multiplexed CPU lines are demultiplexed by the CPU interface circuit to form system bus).

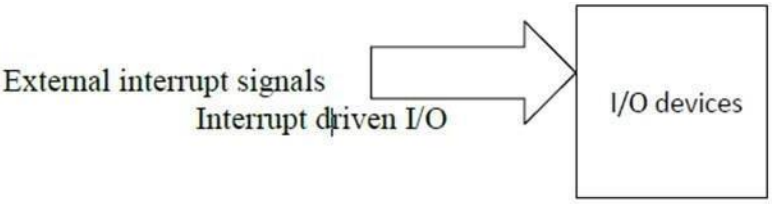
2.1 MEMORY-MAPPING

The memory mapping is the process of interfacing memories to microprocessor and allocating addresses to each memory locations.



Interrupt I/0

If the I/0 device initiate the data transfer through interrupt then the I/0 is called interrupt driven I/0.



EPROM is mapped at the beginning of memory space in 8085 system

In 8085 microprocessor, after a reset, the program counter will have OOOOH address. If the monitor program is stored from this address then after a reset, it will be executed automatically.

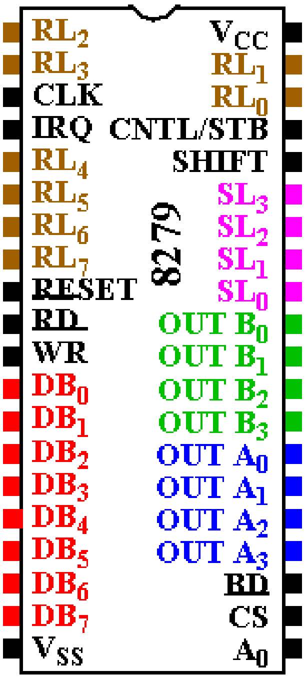
The monitor program is a permanent program and stored in EPROM memory. If EPROM memory is mapped at the beginning of memory space, i.e., at OOOOH, then the monitor program will be executed automatically after a reset. 30. T

The need for system clock and how it is generated in 8085:

The system clock is necessary for synchronizing various internal operations or devices in the microprocessor and to synchronize the microprocessor with other peripherals in the system.

3. 8279PROGRAMMABLE KEYBOARD/DISPLAY INTERFACE

- A programmable keyboard and display interfacing chip.
- Scans and encodes up to a 64-key keyboard.
- Controls up to a 16-digit numerical display.
- Keyboard has a built-in FIFO 8 character buffer.
- The display is controlled from an internal 16x8 RAM that stores the coded display information.



Pinout Definition 8279

- A0: Selects data (0) or control/status (1) for reads and writes between micro and 8279.
- BD: Output that blanks the displays.
- CLK: Used internally for timing. Max is 3 MHz
- CN/ST: Control/strobe, connected to the control key on the keyboard.
- CS: Chip select that enables programming, reading the keyboard, etc.
- DB7-DB0: Consists of bidirectional pins that connect to data bus on micro.

- IRQ: Interrupt request, becomes 1 when a key is pressed, data is available.
- OUT A3-A0/B3-B0: Outputs that sends data to the most significant/least significant nibble of display.
- RD(WR): Connects to micro's IORC or RD signal, reads data/status registers.
- RESET: Connects to system RESET.
- RL7-RL0: Return lines are inputs used to sense key depression in the keyboard matrix.
- Shift: Shift connects to Shift key on keyboard.
- SL3-SL0: Scan line outputs scan both the keyboard and displays.

4. 8255 PROGRAMMABLE PERIPHERAL INTERFACE

Most of µp application will have two processes in common

- Reading data from an external device
- Writing data into another external device.

Special interface circuits are available for this purpose. Such interface circuits are called peripheral interface circuits or I/O ports.

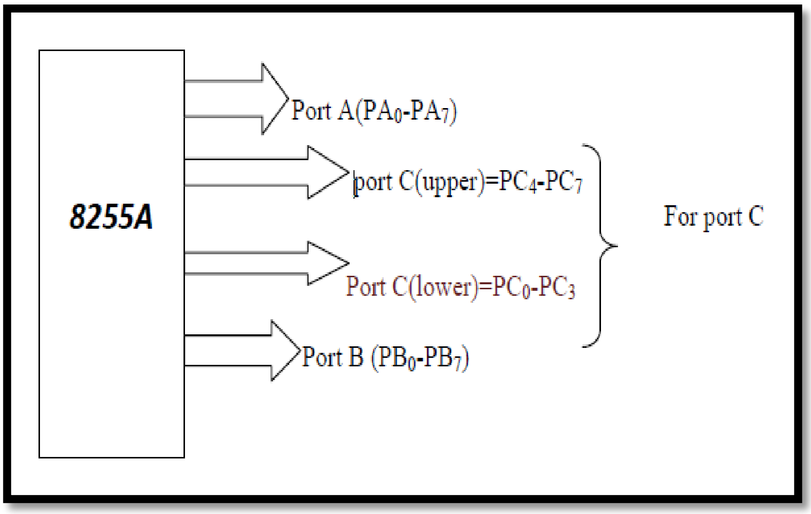
The INTEL 8255 is one such peripheral interface chip. It can be used with INTEL 8085 CPU.

Introduction:

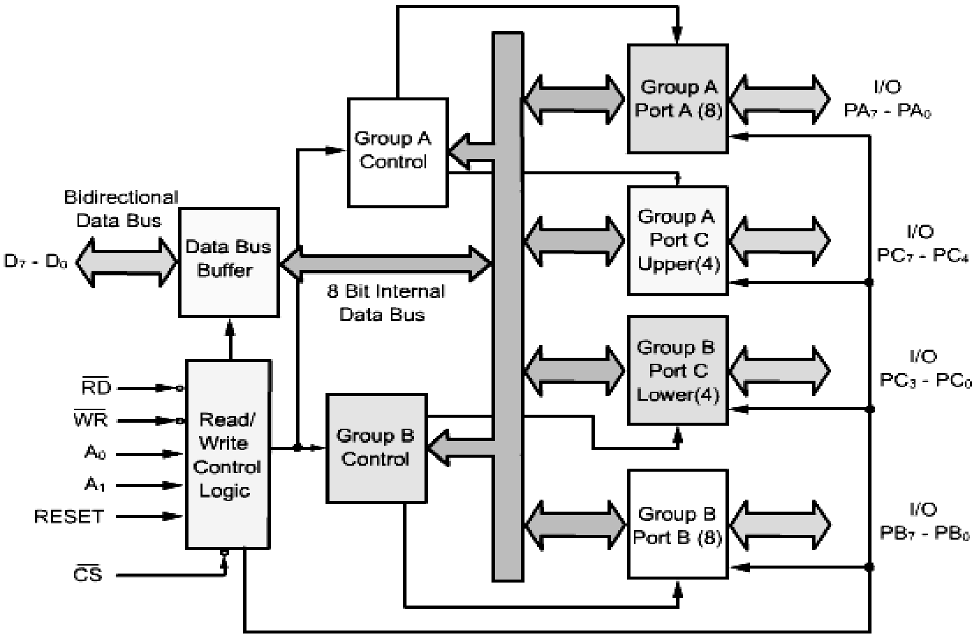
- The 8255A is a widely used programmable, parallel I/O device.
- It can be programmed to transfer data between peripheral devices and µp.
- It is compatible with all INTEL and most other microprocessor.
- It is completely TTL compatible.

Features:

- It is an important general purpose I/O device that can be used with almost any INTEL µp.
- It is packed in 40 pin DIP.
- All I/O pins of pins of 8255 has 2.5 mA DC driving capacity (i.e. sources current of 2.5 mA).
- Out of 40 pins, 24 pins are used as I/O pins. these 24 pins are divided into 3port(A,B,C)
- These ports can be programmed as inputs or outputs.
- The 8 bits of port C can be used as individual bits or be grouped in two 4-bit ports:C upper(cu) and Clower (cl) for the purpose of programming .
- These port are programmable as
    - GROUP A:port A(PA7-PA0) & cu(PC7-PC4)
    - GROUP B:port B(PB7-PB0)& cl (PC3-PC0)



Block diagram



Block diagram of IC 8255

Data bus transfer:

- This tri-state bi-directional buffer is used to interface the internal data bus of 8255 to the system data us.
- Input or output instructions executed by the CPU either read data from or write data into the buffer.
- Output data from the CPU to the port or control register and input data to the CPU from the ports or status register are all passed through the buffer.

Control logic:

- The control logic block accepts control bus signals as well as inputs from the address bus and issues commands to the individual group control blocks (group A control and group B control).
- It issues appropriate enabling signals to access the required data/control words or status word.
- The input pins for the control logic section is described here.

Group A and group B controls:

- Each of the group A and group B control blocks receives control words from the CPU and issues appropriate commands to the ports associated with it.
- The group A control block controls port A and PC7 –PC4 while the group B control block controls port B and (PC3-PC0).
- **Port A:** This has an 8-bit latched and buffered output and an 8-bit input latch. It can be programmed in three modes:
    - Mode-0:simple input/output
    - Mode-1:input/output with handshake
    - Mode-2:bi-directional i/o data transfer
- **Port B:** This has an 8-bit data I/O latch/buffer and an 8-bit data input buffer. it can be programmed in mode 0 and mode 1.
- **Port C:** This has one 8-bit unlatched input buffer and an 8-bit output latch/buffer. it can be separated into two parts and each be used as control signals for ports A and B in the handshake mode. It can be programmed for bit set/reset operation.

Operation modes:

The operation of the ports can be classified into two broad groups

- I/O mode and
- Bit set/reset mode.

Bit set-reset (BSR) mode:

- The individual bits of port C can be set or reset by sending out a single OUT instruction ti the control register.
- When port C is used for control/status operation, this feature can be used to set or reset individuals bit.

I/O modes:

Mode 0:

- In this mode ports A and B are used as two simple 8-bit I/O ports and ports C as two 4-bit ports.
- Each port can be programmed to function as simply an input port or output port.

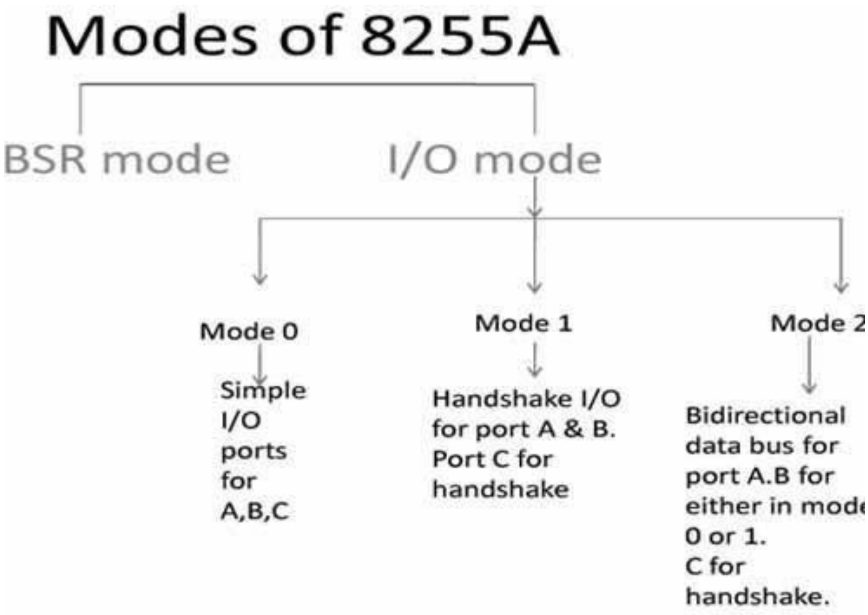
- The input/output features in mode 0 are as follows
  - Outputs are latched
  - Inputs are buffered ,not latched
  - Ports do not have handshake or interrupt capability.

Mode 1:

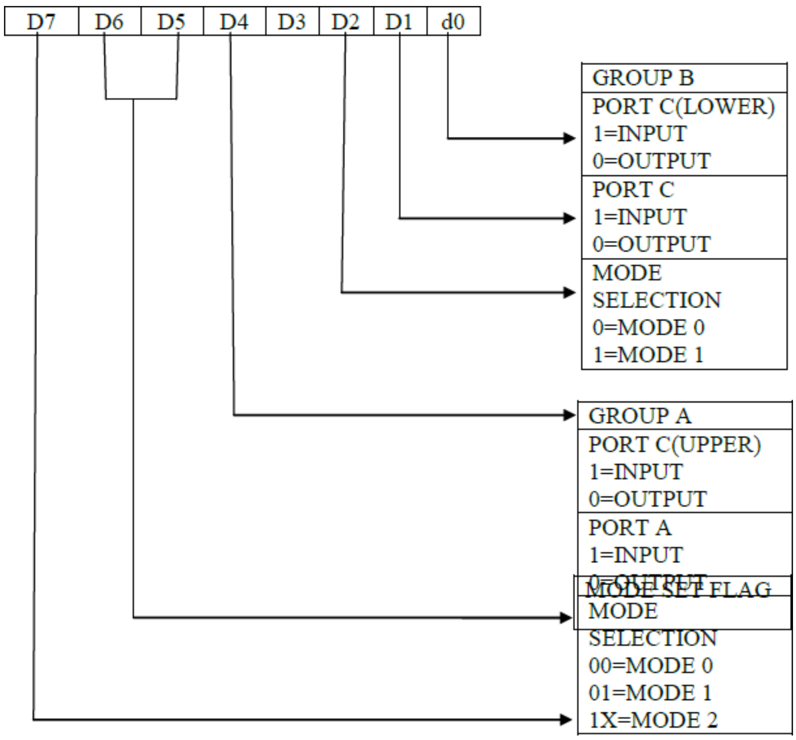
- Under mode 1 the port A or port B can be set for input or output operation.
- The bits from port C are used as control signals.
- These signals are used for handshaking.
- This supports handshaking has following features.
  - Port A & b function as 8-bit I/O ports. they can be configured either as inputs or output ports.
  - Each port uses three lines from port C as handshake signals.
  - The remaining two lines of port C can be used for simple I/O functions.
  - Input and output data are latched.
  - Interrupt logic is supported.

Mode 2:

- In mode 3 the port A is used as a bi-directional port with simultaneous input and output capability.
- Port C is used for handshake.



CONTROL WORD



Control section:

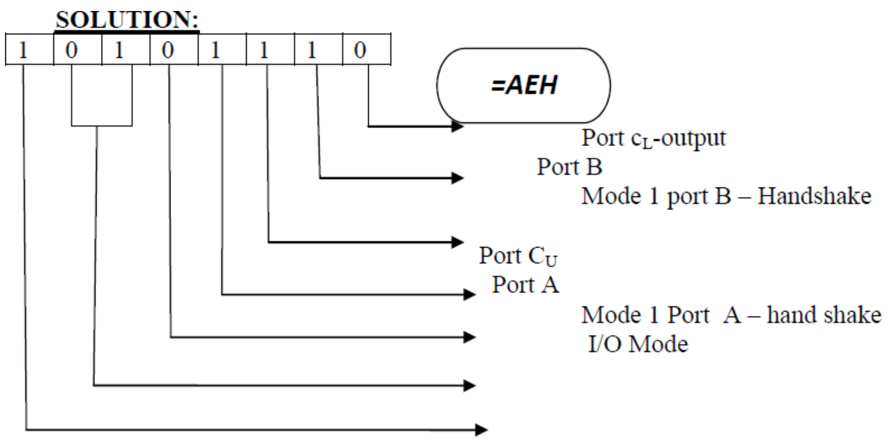
- $\overline{RD}$, $\overline{WR}$, ¯ ¯ RESET, are 4-control signals generated by control section. The function of these control lines are $\overline{RD}$ (Read bar)=>A low on this control lines enables the microprocessor to read data from selected I/O ports of the 8255.
- $\overline{WR}$(Write bar) => A low on this line enables the microprocessor to writes (loads) operation of data into the selected I/O ports or the control register of the 8255.
- RESET(Reset) =>A high on this line clears the control register and sets the ports in the input mode.
- $\overline{CS}$(Chip select bar)=>A high on this pin selects 8255 chip. With the high signal the data bus buffer that connects the 8255 to the system data bus remains floated.
- The source of the $\overline{CS}$ signal depends on whether isolated or memory mapped I/O is used. For isolated I/O, bits AD2□AD7 may be decoded to provide $\overline{CS}$, and its bits AD1 and AD8 may be used for selecting the control register or any of the three ports. If the six address bits AD2□AD7 are decoded, as many as 2^6(=64) 8255 can be selected in any system. With isolated I/O, selection of the 8255 is further conditioned on IO/$\overline{M}$ =1. Linear selection saves decoders and can select 6-numbers of 8255, using isolated I/O with device and port selection.

Programming example:

1. Write a program to initialize 8255 in the configuration given below:

1. Port A: Output with handshake
2. Port B: Input with handshake
3. Port CL: output
4. Port CU: input

Assume address of the control word register of 8255 as 23H.

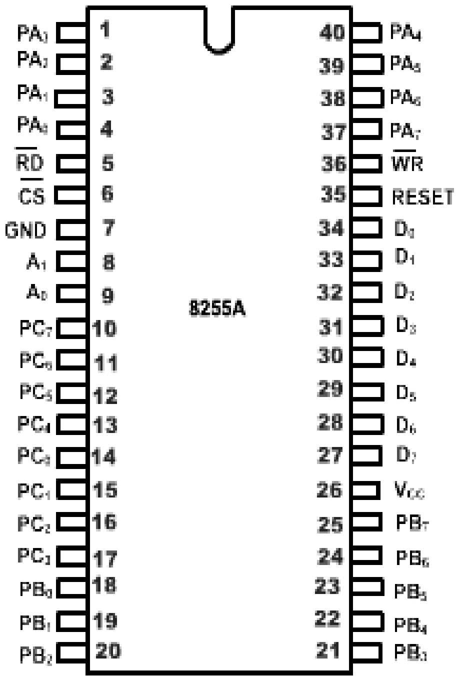


Source program:

```
MVI A, AEH;        LOAD CONTROL WORD
OUT 23H;           SEND CONTROL WORD
```

PIN DIAGRAM:

Pin Configuration



Pin Names

D_7 - D_0	Data Bus (Bidirectional)
RESET	Read Input
$\overline{CS}$	Chip Select
$\overline{RD}$	Read Input
$\overline{WR}$	Write Input
A_0, A_1	Port Address
PA_7 - PA_0	Port A (Bit)
PB_7 - PB_0	Port B (Bit)
PC_7 -PC_0	Port C (Bit)
V_{CC}	+ 5 volts
GND	0 volts

Pin symbols	function
D0-D7(data bus)	Bi-directional,tri-state data bus connected to system data bus used to transfer data and control word from µp(8085) to 8255 or to receive data or status.
PA0-PA7(port A)	8-bit Bi-directional I/O pins used to send data to output device and to receive data from input device.
PB0-PB7(port B)	8-bit Bi-directional I/O pins used to send data to output device and to receive data from input device
PC0-PC7	8-bit Bi-directional I/O pins divided into 2groups $PC_L(PC_3-PC_0)$and $PC_U(PC_7-PC_4)$
$\overline{rd}$(read)	When the pin is low,the cpu can read the data in the data in the ports or the status word through buffer.
$\overline{WR}$(write)	When the input pin is low,the cpu can write the data in the data on the ports or in control register through the data buffer.
$\overline{CS}$(chip select)	This is an active low input which can be enabled for data transfer operation b/w CPU and the 8255.
RESET	This is an active high input used to reset 8255.
A_0 and A_1	These along with RD and WR inputs control selection of the control/status registers or one of the three ports.

A_1	A_0	$\overline{RD}$	$\overline{WR}$	$\overline{CS}$	operation
					INPUT (READ) OPERATION
0	0	0	1	0	Port A to data bus
0	1	0	1	0	Port B to data bus
1	0	0	1	0	Port C to data bus
					Output(write) operation
0	0	1	0	0	Data bus to port A
0	1	1	0	0	Data bus to port B
1	0	1	0	0	Data bus to port C
1	1	1	0	0	Data bus to control register
					Disable function
x	x	x	x	1	Data bus tri -stated
1	1	0	1	0	Illegal condition
x	x	1	1	0	Data bus tri-stated

Applications:

- **Printer operation:**

The data transfer operation between a printer and CPU can be carried out as explained below:

1. The CPU places an 8 bit data in the output port.

2. The operation circuit sends out OBF (output buffer full), active low level signal to the data is available for taking inside the printer memory buffer.

3. If the printer buffer is empty enough to receive the data, it will read the daa into the pointer buffer and if it is ready to receive next byte of data, then it will acknowledge the output port through the ACK active low signal.

4. The active ACK level is sensed by the interface circuit and an interrupt is raised to inform the CPU to place the next data from the computer memory to the output port.

5. Steps 1, 2, 3 and 4 are replaced.

This operation is continued till the active ACK signal is received from the printer.

5. CONCEPT OF SERIAL COMMUNICATION INTERFACE

INTRODUCTION:

Transmitting data over a long distance, using parallel communication is impractical due to increase in cost cabling. Parallel communication is also not practical for devices such as cassette tapes or a CTR terminal. In such situations, serial communication is used. In serial communication one bit is transferred at a time over a single line.

CLASSIFICATION:

Serial data transmission can be classified on the basis of how transmission occurs

1. Simplex
2. Half duplex
3. Full duplex

Simplex:

In simplex the hardware exists such that data transferred takes place only in one direction. There is no possibility of data transfer in the other direction. A typical example is transmission from a computer to the printer.

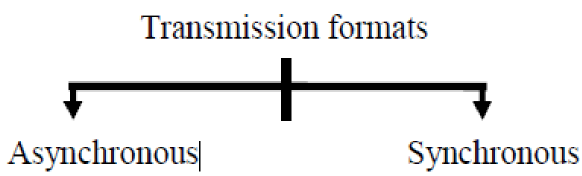
Half duplex:

The half duplex transmission allows the data transfer in both direction but not simultaneously. A typical example is a walkie-talkie

Full duplex:

The full duplex transmission allows the data transfer in both directions simultaneously. A typical example is transmission through telephone lines.

TRANSMISSION FORMATS:



Asynchronous:

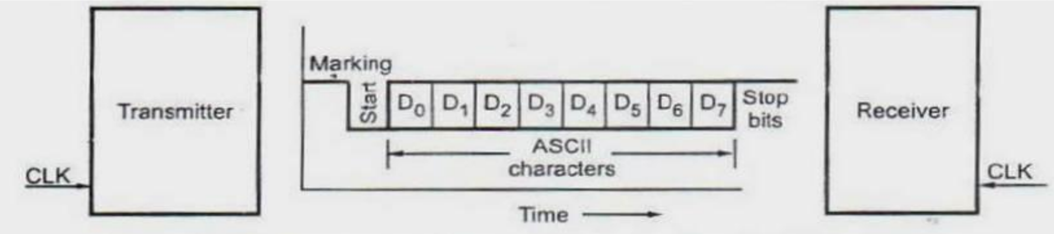
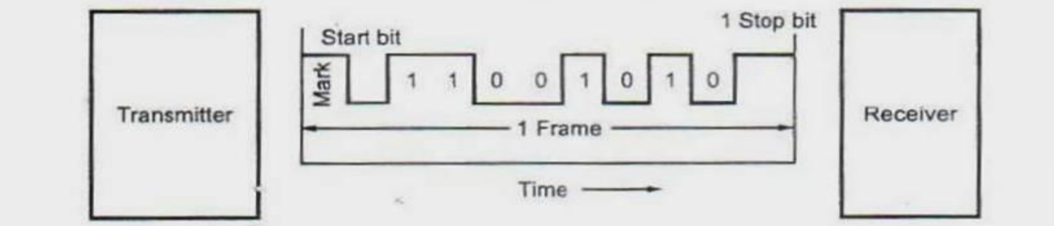


Fig. 11.1 Transmission format for asynchronous transmission

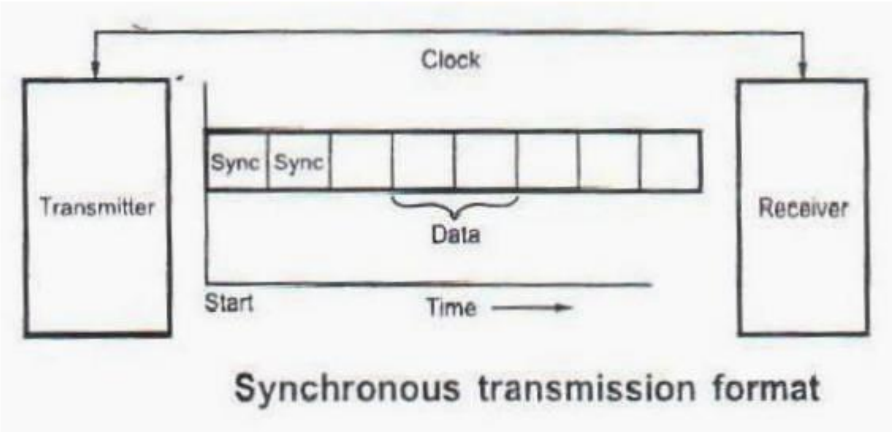


Asynchronous format with data byte CAH

- Asynchronous formats are character oriented. In this, the bits of a character or data word are sent at a constant rate, but characters can come at any rate as long as they do not overlap.
- When no characters are being sent, a line stays high at logic 1 called mark, logic 0 is called space.
- The beginning of a character is indicated by a start bit which is always low. This is used to synchronize the transmitter and receivers. After the start bit, the data bit are sending with least significant bit first, followed by one or more start bits.
- The stop indicates the end of character. Different systems use 1, 1 1/2 or 2 stop bits. The combination of start bits, character and stop bits is known as frame.
- Start and stop bits carry no information, but are required because of asynchronous nature.
- The data byte CAH would look when transmitted in the asynchronous serial format.
- The data rate can be expressed as bits/ sec. or characters/sec. the term bits/sec. is also called baud rate. Asynchronous format is generally used in low speed transmission (< 20 Kbits/sec)

Synchronous:

- The start and the stop bits in each frame of asynchronous format represents wasted overhead bits that reduce the overall character rate.
- The start and stop bits can be eliminated by synchronizing receivers and transmitters. They can be synchronized by having a common clock signal.
- Such communication is called **synchronous serial communication.**
- In the synchronous serial communication synchronous bits are inserted instead of start and stop bits.



Comparison between Asynchronous And Synchronous Serial Data Transfer:

S. No	Asynchronous serial communication	Synchronous serial communication
1	Transmitters and receivers are not synchronized by clock	Transmitter and receivers are synchronized by clock
2	Bits of data are transmitted at constant rate	Data bits are transmitted with synchronization of clock
3	Character may arrive at any rate at receiver	Character is received at constant rate
4	Data transfer is character oriented	Data transfer takes place in blocks
5	Start and stop bits are required to establish communication of each character	Start and stop bits are not required to establish communication of each character, however, synchronization bits are required to transfer the data block
6	Used in low-speed transmissions at about speed less than 20 Kbits/sec	Used in high-speed transmissions

6. USART-8251A

- To transmit byte data it is necessary to convert into eight serial bits. This can be done by using the parallel to serial converter.
- Similarly at the reception the serial bits must be converted into parallel 8 bit data. The serial to parallel converter is used to convert serial data bits into the parallel data.
- The devices are designed for these purposes are called Universal Asynchronous Receivers- Transmitter.
- A good example of UART is 8250 and USART is 8251. These devices are software programmable for numbers of data bits parity and number of stop bits.

Features:

1. The Intel 8251A is a universal synchronous and asynchronous communication controller.

2. It supports standard asynchronous protocol with:

a) 5 to 8 Bit character format

b) Odd, even or not parity generation and detection.

c) Baud rate from DC to 19.2 Kbaud

d) False start bit detection.

e) Automatic break detect and handling.

f) Break character generation

3. It has built in baud rate generator

4. It supports standard synchronous protocol with:

   a) 5 to 8 Bit character format

   b) Internal or external character synchronization

   c) Automatic sync insertion

   d) Baud rate from DC to 64 kbaud

5. It allows full duplex transmission and reception

6. It provides double buffering of data both in the transmission section and in the receiver section.

7. It provides error detection logic, which detects parity, overrun and in the receiver section.

8. It has Modern Control Logic, which support basic data set control signals.

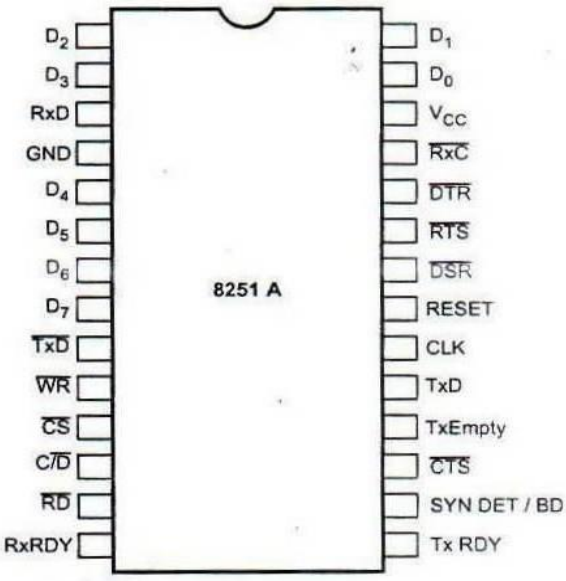
9. It provides separate clock inputs for receiver and transmitter section, thus providing an option of fixing different baud rates for the transmitter and receiver section.

10. It is compatible with an extended range of Intel microprocessors

11. It is fabricated in 28 pins DIP package and its all inputs and outputs are TTL compatible.

12. It is available in standard as well as extended temperature range.

PIN DIAGRAM OF 8251A



Pin diagram of 8251A

Data Bus: Bi-directional, tri-state, 8-bit Data Bus. This pin allow transfer of bytes between the CPU and the 8251A

RD (Read): A low on this input allows the CPU to read data or status bytes from 8251A

WR (Write): A low on this input allows the CPU to write data or command word to the from 8251A

CLK (Clock): The CLK input is used to generate internal device timing. The frequency of CLK must be greater than 30 times the receiver or transmitter data bit rates.

RESET: A high on this input forces the 8251A into an “Idle” mode. The device will remain at “Idle” until a new set of control words is written into the 8251A to program its functional definition.

C/D (Control/Data): This input in conjuction with the WR and RD inputs, informs the 8251A that the word on the Data Bus is either a data character, control word or status information.

CS (Chip Select): A low on this input allows communication between CPU and 8251A

C/D	RD	WR	OPERATION
0	0	1	CPU reads data from USART
0	1	0	CPU writes data to USART
1	0	1	CPU reads status from USART
1	1	0	CPU writes command to USART
X	1	1	USART Bus floating

MODERN CONTROL SIGNALS:

The 8251A has a set of control inputs and outputs that can be used to simplify the interface to almost any modem.

DSR (Data Set Ready): This input signal is used to test modem conditions such as data set ready.

DTR (Data Terminal Ready): This output signal is used to tell modem that data terminal is ready.

RTS (Request To Send): This output signal is asserted to begin transmission.

CTS (CLEAR TO SEND): A low on this input enables the 8251A transmit serial data if the TxE bit in the command byte is set to a "one".

TRANSMITTER SIGNALS:

TxD: Transmit Data: This output signal outputs a composite serial stream of data on the falling edge of TxC

TxRDY (Transmitter Ready) : This output signal indicates the CPU that the transmitter is ready to accept the data character

TxE (Transmitter Empty*):* This output signal indicates that the transmitter has no character to transmit.

TxC (Transmitter Clock): This clock input controls the rate at which the character is to be transmitted

RECEIVER SIGNALS:

RxD (Receiver Data): This input receives a composite serial stream of data on the rising edge of RxC

RxRDY (Receiver Ready): This output indicates that the 8251A contains a character that is ready to be input to the CPU

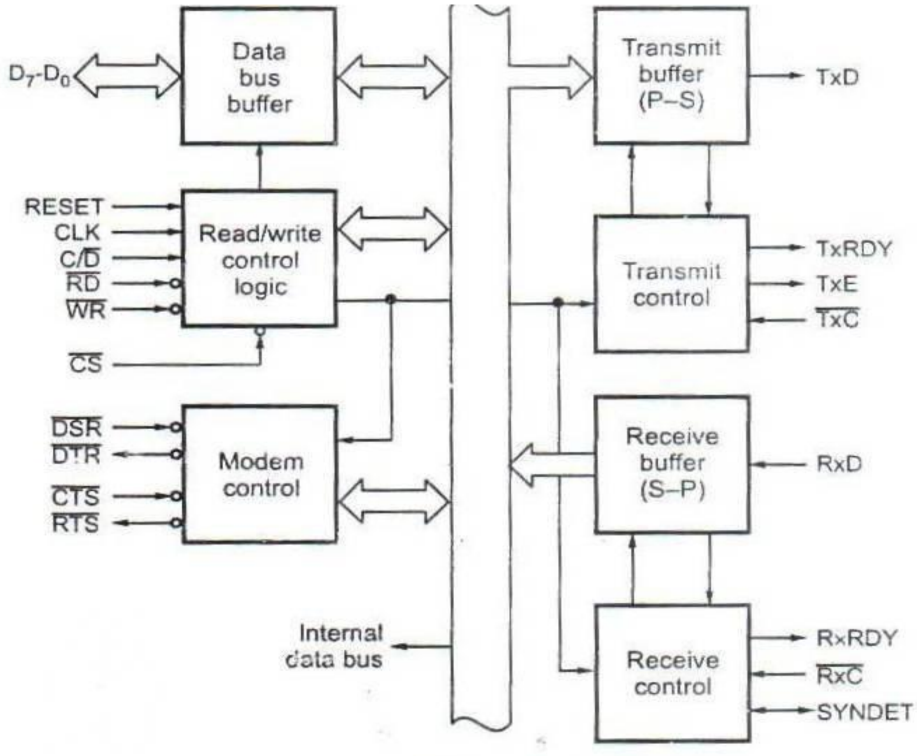
RxC (Receiver Clock): This clock input controls the rate at which the character is to be received.

SYNDET (Sync Detect) / BRKDET (Break detect)

- This pin is used in synchronous mode for detection of synchronous characters and may be used as either input or output
- In asynchronous mode this pin goes high if receiver line stays low for more than 2 character times.
- It then indicates a break in the data stream.

- When used as an input (external sync detect mode) a positive signal will cause the 8251A to start receiving data characters on the rising edge of the next RXC

BLOCK DIAGRAM



Block diagram

Data Bus Buffer

- This tri-state, bi-directional, 8-bit buffer is used to interface 8251 to the system data bus.
- Along with the data, control word, command words and status information are also transferred through the Data Bus Buffer.

Read/ write control logic:

- This functional block accepts input from the system control bus and generates control signal for overall device operation
- It decodes control signals on the 8085 control bus into signal which controls the internal and external I/O bus.
- It contains the control word register and command word register that stores the various control formats for the device functional definition

Transmit Buffer

- The transmit buffer accepts parallel data from the CPU, adds the appropriate framing information, serializes it, and transmits it on the TxD pin on the falling edge of TxC
- It has two registers
- A buffer register to hold eight bits and an output register to convert eight bits into a stream of serial bits
- The CPU writes a bit in the buffer register, which is transferred to the output register when it is empty.
- The output register then transmits serial data on the TxD pin
- In the asynchronous mode the transmitter always adds START bit; depending on how the unit is programmed, it also adds an optional even or odd parity bit, and either 1, 1 ½ , or 2 STOP bits.
- In synchronous mode no extra bits (other than parity, if enable) are generated by the transmitter.

Transmit control

- It manages all activities associated with the transmission of serial data
- It accepts and issues signals both externally and internally to accomplish this functions

TxRDY (Transmit Ready)

- This output signal indicates CPU that buffer register is empty and the USART is ready to accept a data character
- It can be used as an interrupt to the system or, for polled operation, the CPU can check TxRDY using the status read operation
- This signal is reset when a data bit is loaded into the buffer register

TxE (Transmitter Empty*)*

- This is an output signal
- A high on this line indicates that the output buffer is empty
- In the synchronous mode, if the CPU has failed to load a new character in time, TxE will go high momentarily as SYN characters are loaded into the transmitter to fill the gap in transmission

TxC (Transmitter Clock)

- This clock controls the rate at which characters are transmitted by USART
- In the synchronous mode TxC is equivalent to the baud rate, and is supplied by modem
- In the asynchronous mode TxC is 1, 16,or 64 times baud rate
- The clock division is programmable
- It can be programmed by writing proper mode word in the mode set register

Receiver Buffer

- The receiver accepts serial data on the RxD line, converts this serial data to parallel format, check for bits or characters that are unique to the communication technique and sends an "assembled" character to the CPU
- When 8251A is in the asynchronous mode and it is ready to accept a character, it looks for a low level RxD line.
- When it receives the low level, it assumes that it is a START bit and enables an internal counter.
- At a count equivalent to one-half of a bit time, the RxD lines sampled again
- If the line is still low a valid START bit is detected and the 8251A proceeds to assemble the character
- After successful reception of a START bit the 8251A receives data, parity and STOP bits, and then transfers the data on the receivers input register
- The data is then transferred into the receiver buffer register
- In the synchronous mode the receiver simply receives the specified number of data bits and transfers them to the receiver input register and then to the receiver buffer register

Receiver Buffer

- It manages all receiver-related activities.
- Along with data reception , it does false start bit detection, parity error detection, framing error detection, sync detection and break detection

RxRDY (Receiver Ready)

- This is an output signal
- It goes high(active) when the USART has a character in the buffer register and is ready to transfer it to the CPU
- This line can be used either to indicate the status in the status register or to interrupt the CPU
- This signal is reset when a data bit from receiver buffer is read by the CPU

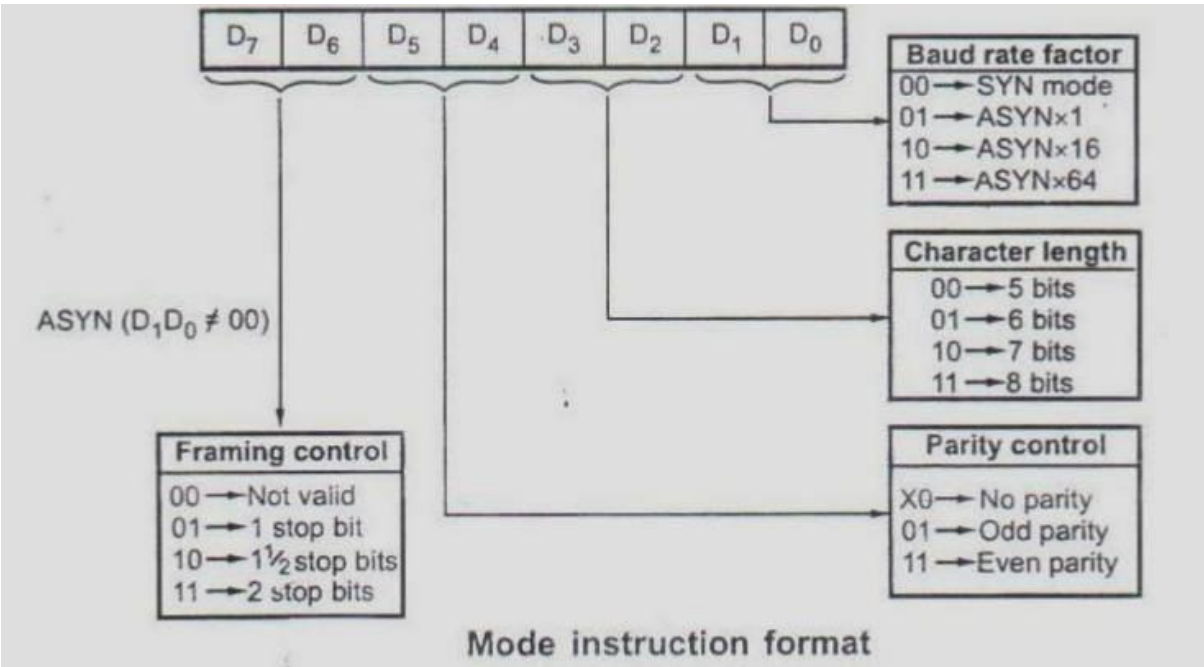
RxC (Receiver Clock)

- This clock controls the rate at which characters is to be received by USART
- In the synchronous mode RxC is equivalent to the baud rate, and is supplied by modem
- In the asynchronous mode RxC is 1, 16,or 64 times baud rate
- The clock division is programmable
- It can be programmed by writing proper mode word in the mode set register

Modem control

- The 8251 has a set of control inputs and outputs that can be used to simplify the interface to almost any modem
- It provides control circuitry for the generation of RTS and DTR and the reception of CTS and DSR
- A general purpose inverted output and general purpose input are provided
- The output is labelled DTR and the input is labelled DSR
- DTR can be asserted by setting bit 2 of the command instruction;
- DSR can be sensed as bit 7 of the status register
- When used as a modem control signal DTR indicates that the terminal is ready to communicate and DSR indicates that it is ready for communication

8251A CONTROL WORDS



- The control words defines the complete functional definition of 8251A and they must be loaded before transmission or reception
- The control words of 8251A are split into two formats:

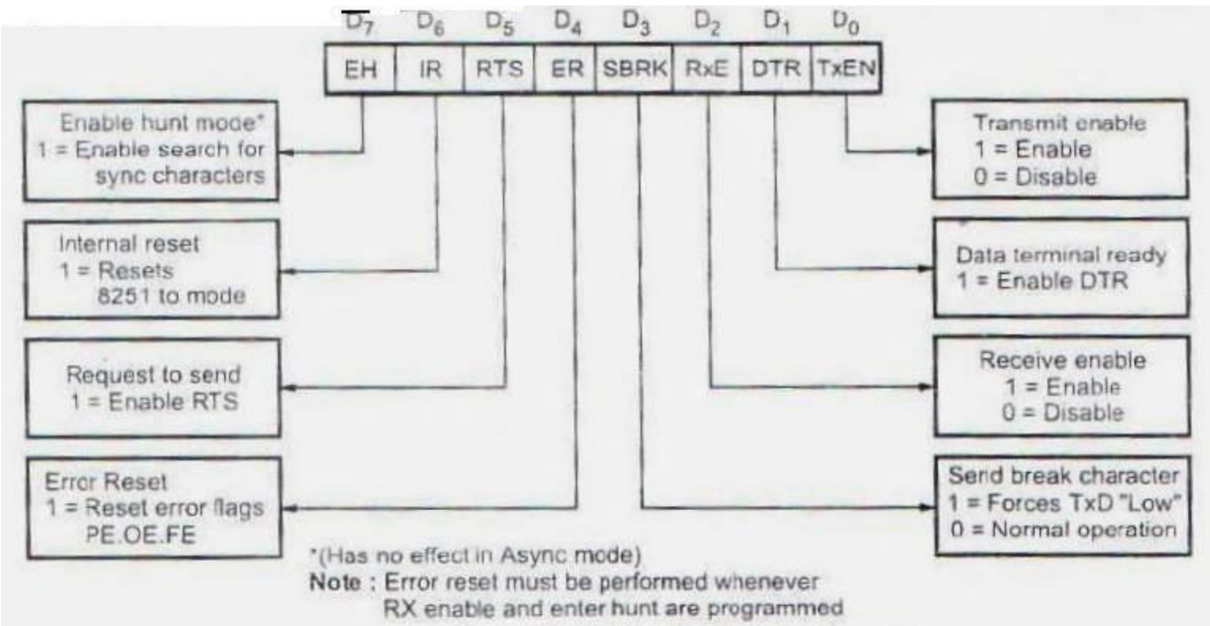
  A. Mode instruction

  B. Command instruction

- The instruction can be considered as four 2 –bit fields
- The first 2-bit field (D1-D0) determines whether the USART is to operate in the synchronous (00) or asynchronous mode.

- In asynchronous mode, this fields determines the division factor for clock to decide the baud rate.
- For example, if D1 and D0 are both ones, the RxC and TxC will be divided by 64 to establish the baud rate.
- The second 2-bit field (D3-D2) determines number of data bits in one character.
- With this 2-bit field we can set character length from 5-bits to 8-bits
- The third 2-bit field, (D5 – D4) controls the parity generation. The parity bit is added to the data bits only if parity is enabled.
- The last field, (D7 – D6), has two meanings depending on whether operation is to be in the synchronous or asynchronous mode, (i.e. D1 D0 not equal to 00
- It controls the number of STOP bits to be transmitted with the character
- In synchronous mode, (D1 D0 = 00) this field controls the synchronizing process.
- It decides whether to operate with external synchronization or internal synchronization and whether to transmit single synchronizing character or two synchronizing characters

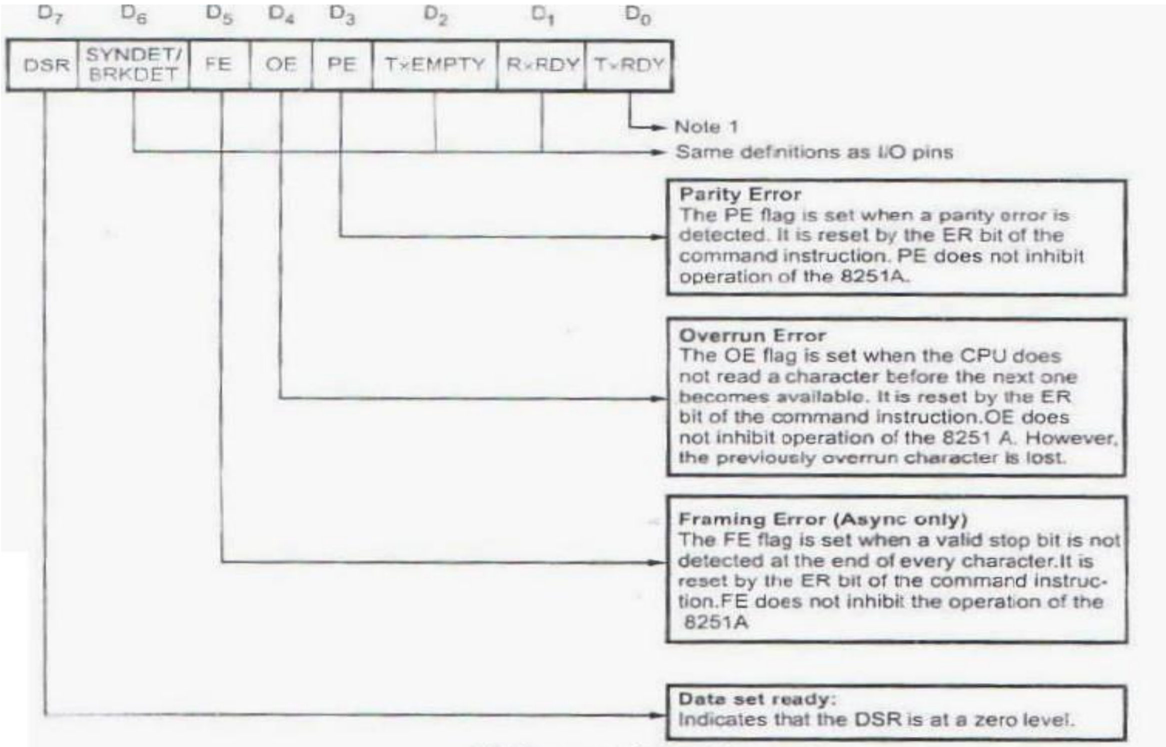
COMMAND INSTRUCTION



Command instruction format

- After the mode instruction, command character should be issued to the USART.
- It controls the operation of the USART within the basic frame work established by the mode instruction
- It does function such as: enable transmit /receive, error reset and modem control

8251A STATUS WORDS



Status register format

- In the data communication systems it is often necessary to examine the “status” of the transmitter and receiver
- It is also necessary for CPU to know if any error has occurred during communication

Line Pin	Description	E/A Destinations
1	Protective ground	AA
2	Transmitted data	BA
3	Received data	BB
4	Request to send (RTS)	
5	Clear to send (CTS)	CA
6	Data set ready (DSR)	CB
7	Signal ground (common return)	AB
8	Received line signal detector (data carrier detect)	CF
9	Reserved for data set testing	
10	Reserved for data set testing	
11	Unassigned	
12	Secondary received line signal detector	SCF
13	Secondary clear to send	SCB
14	Secondary transmitted data	SCA
15	Transmissional signal element timing (CDE source)	DB
16	Secondary received data	SBB
17	Receiver signal element timing (DOE source)	DD
18	Unassinged	
19	Secondary request to send	SCA
20	Data terminal ready	CD
21	Signal quality detector	CG
22	Ring indicator (R)	CE
23	Data signal rate selector (DTE/DCE source)	CA/CI
24	Transmit signal element timing (DTE source)	DA
25	Unassigned	

- The 8251A allow the programmer to read above mentioned information from the status register any time during the functional operation

ERROR DEFINITIONS

Parity Error: At the time of transmission of data an even or odd parity bit is inserted in the data stream. At the receiver end, if parity of the character does not match with the pre-defined parity, parity error occurs.

Overrun Error: In the receiver section received character is stored in the receiver buffer. The CPU is supposed to read this character before reception of the next character. But if CPU fails in reading the character loaded in the receiver buffer, the next the received character replaces the previous one and the OVERRUN error occurs

Framing Error: If valid stop bit is not detected at the end each character framing error occurs.

All these errors, when occur, set the corresponding bits in the status register. These error bits are reset by setting ER bit in the command instruction.

7. RS-232 INTERFACE

- The RS-232 standard defined a scheme for asynchronous communication, where there is a specified timing between data bits no fixed timing between the characters that the bits form
- In contrast synchronous communication requires specific timing between bits and character, and the clock for recovering data bits must be synchronized to or derived from the bit stream.
- Asynchronous system are simpler but transmit data at much lower rates and are therefore less efficient than synchronous system
- Communication defined by RS-232 is serial data communication
- There is a single wire or link for each direction of data flow and the bits of the message are sent in sequence one at a time
- Compared to parallel systems where multiple lines carry many data bites simultaneously, serial communication requires less circuitry at either end of the data link, is serial system have lower through put than parallel systems, but in practice in parallel link is practical only over a distances of a few feet
- All though RS-232 standard defines a serial system with just a single wire for each direction , the standard allows for other signal wires the DTE and DCE as well
- These additional signals are used to manage the data interface and to indicate communication status at any time to both the DTE device and the DCE device
- The RS-232 specification is intended to provide reliable communication up to distances of soft, at rate up to 20000 bands.
- However RS -232 interconnection should not be longer than soft, it has been successfully used for longer distance
- Since the limiting factors of capable capacitance and noise are much less critical when lower band values are used ; there are successful RS-232 links several hundred feet long , although this exceeds what the standard is designed to allow
- Finally the phrase serial ASCII format is often used in conjunction with RS-232 .
- The format standard itself does not specify at all what bit patterns should be used to represent the actual information being transmitted
- In special circumstances ,however non-ASCII format are used ,especially where one number is being sent

- As we saw in the discussion of ASCII , it is relatively inefficient for numerical information since it requires a seven or eight bit field for each digit of the entire number
- In contrast a straight binary format can express values up to 255 in an eight bit field
- However, unless noted otherwise, we will assume that the RS-232 data are in ASCII format.
- RS -232 is primarily a signal voltage and timing ,DTE to DCE interface standard
- Just as it does not discuss how the data should be represented by the bits ,it does not define the overall message format and protocol
- The definition of these are left to the developers of the communication system
- This means that RS-232 provides a great deal of flexibility, but also can have problems when signals levels and timing agree but message format, content and protocol differ between the two DTE devices that in inter connects in a complete system
- The RS-232 standard defined twenty five signal lines
- This does not mean that every DTE to DCE interface requires 25 signal wires many RS-232 interface use just a few
- The lines are divided into four group: data , control , timing and specifies ground for protection and to make sure that the DCE and DTE chasis are at the same electric potential
- In practice the signal group and chasis gnd are often the same wire
- Are chasis ground is omitted ,which are violation of the standard requirement; fortunately ,this usually works, but it can cause operational(or even safely) problems
- The data line are most important signals
- Received data and transmitted data line permits full-duplex communication between the DTE and DCE
- To make sure that there is no confusion on the direction of the data flow on the received data and transmitted data line, the RS-232 standard specifies data flow direction from the perspective of the DTE: data go from the DTE to DCE on transmit data line and data go from the DCE to DTE and the reserved data line from the perspective of the DCE, then data are transmitted on what is called the received data line and received on what is called transmitted data line

- This may seen confusing but it really avoids the more serious problem that will occur if both units try to transmit on the same time (single clash) or receive on the same wire(there is no signal present, but there are two listeners on the same line and no signal source)
- These lines are used for handshaking ,so either the DTE or the DCE can signal to the other that there are data to be transmitted and can indicate to the other if either DTE or DCE is ready to accept new data
- Four control lines are used most frequently

    1. Request to send (RTS) from the DTE signals the DCE that the DTE has new data it would like to transfer

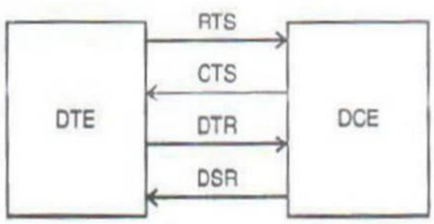
    2. Clear to send (CTS) from the DCE indicates to the DTE that the DCE can accept new data

    3. Data terminal ready (DTR) from the DTE is another indication that the DTE is ready

- Note that in the RS-232 standard the hand shake lines are asserted or active(binary value=1) when they are in the space zone of +3 to 3 to +25 V, which also corresponds to a binary zero for data
- This sometimes causes confusion when trouble shooting since an asserted control line is said to be in binary 1 state , yet its signal voltage is the same as a data bit that is binary 0.
- As long as everyone follows the standard , however there is no operating signal
- Using these handshaking lines the complete data transmission sequence is a series of repeated request to send new data, a handshake showing that the request can be honoured, sending the data ,a hand shake indication that new data cannot be accepted because either the DCE or DTE is full or the transmission link is unavailable, then watching for the ‘OK to send’ signal finally appearing and so on
- The other control lines let the DCE indicate to the DTE it has detected a ringing signal or that the communication lines signal quality is acceptable among other function

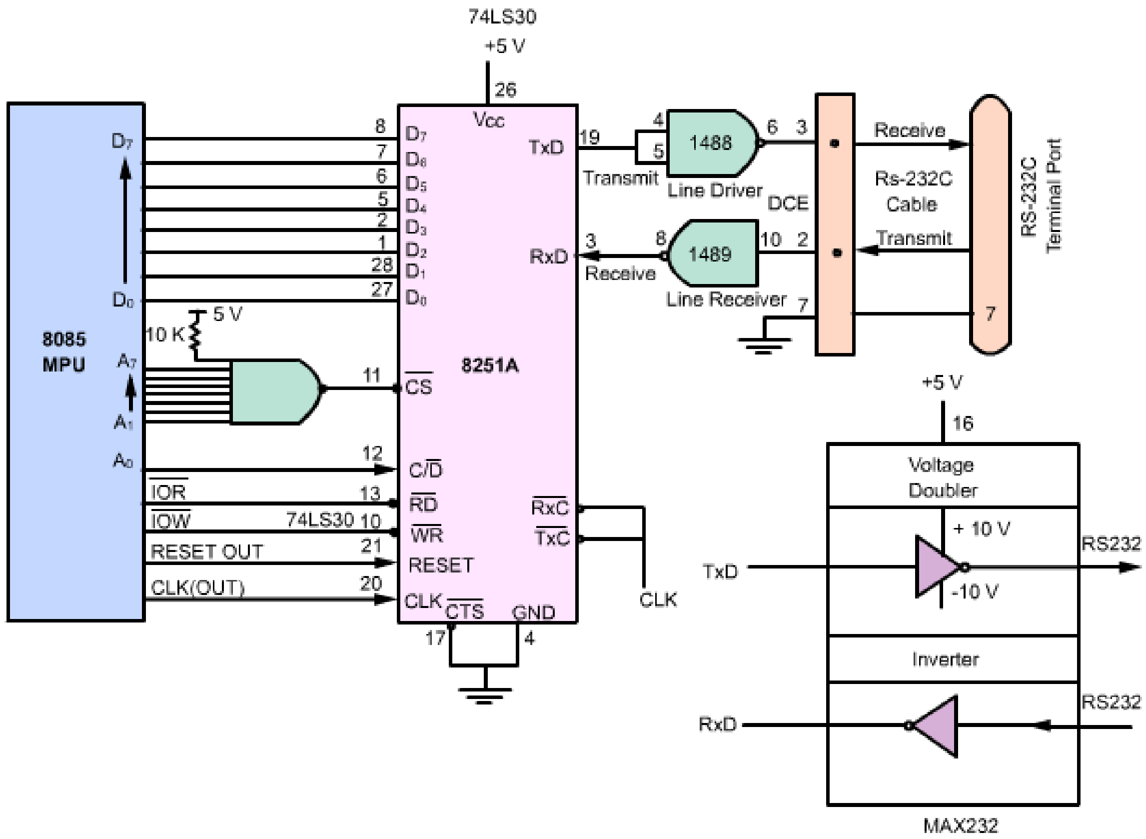
	Line Pin	Description	Gnd	Data From DCE	Data To DCE	Control From DCE	Control To DCE	Timing From DCE	Timing To DCE
Data	1	Protective ground	×						
	7	Signal ground/common return	×						
Control	2	Transmitted data			×				
	3	Received data		×					
	4	Request to send (RTS)					×		
	5	Clear to send (CTS)				×			
	6	Data set Ready (DSR)				×			
	20	Date-terminal ready (DTR)					×		
	22	Ring detector				×			
	8	Received line signal detector or data carrier detect (DCD)				×			
	21	Signal quality detector				×			
	23	Data signal rate selector (DTE)					×		
	23	Data signal rate selector (DCE)				×			
Timing	24	Transmitter signal element timing (DTE)							×
	15	Transmitter signal element timing (DCE)						×	
	17	Receiver signal element timing (DCE)						×	
Secondary group	14	Secondary transmitted data			×				
	16	Secondary received data		×					
	19	Secondary request to send					×		
	13	Secondary clear to send				×			
	12	Secondary received time signal detector				×			

RS-232 lines, organized by functional grouping together with direction of signal flow

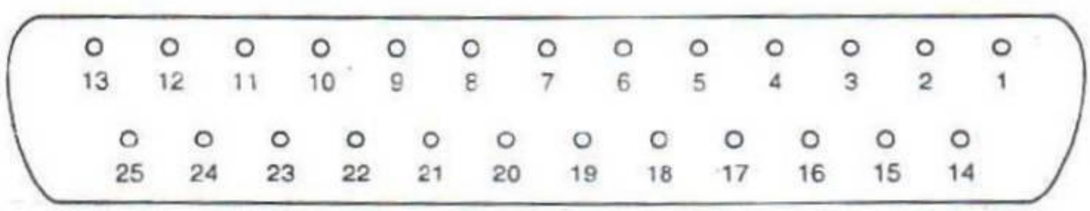


Four key control (handshaking) lines for DTE/DCE interfacing

RS-232 CONNECTORS



- For many years, a 25 pin D-shaped connector with 13 pins, on one row and 12 on second row was most common connector for RS-232.
- However, since many of the 25 wires specify by the standard are not used, this is wasteful and costly in terms of wire termination and circuit board space.
- Newer systems often use a smaller connector such as a nine pin D- shaped unit, or even a smaller type with just a few wires at the DTE end of the cable.
- However most DCE devices, such as printers, terminals or modems, continue to use 25 pin –shaped connector, although this is changing.



. Pin diagram of 25 pin connector used for interfacing